

Low rank is enough for the neural tangent kernel: depth–rank tradeoff in neural network optimization

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Editor: Under Review for COLT 2026

Abstract

We study low-rank random-feature networks (RF-LR), where each layer is a frozen random feature map followed by a trainable readout into a bottleneck of dimension $r \ll N$. We focus on a finite-sample kernel regime: we take a sequential infinite-width limit $N \rightarrow \infty$, and then study how the remaining bottleneck randomness concentrates as $r \rightarrow \infty$.

Our main results quantify how depth L and bottleneck dimension r affect conditioning for kernel regression in the lazy/NTK regime. We derive an explicit RF-LR NTK recursion (with an explicit $1/r$ bottleneck factor) and analyze the deterministic proxy mean recursion: the kernel magnitude saturates with depth while the diagonal–off-diagonal gap decays. All condition-number bounds in this paper refer to the *proxy* Gram matrix. We prove a *lower bound* $\kappa \geq \Omega(r \cdot L)$ in general; in one special case—equicorrelated data or high-dimensional random data (pairwise correlations $O(1/\sqrt{d})$ w.h.p.)—we give one explicit result with exact conditioning ($\kappa_{\perp} = 1$ or $1 + o(1)$) and full proof in the appendix. We do not claim that these bounds govern optimization. We also show that *low rank is enough for the NTK RKHS in the three-layer case*: the mean three-layer RF-LR kernel induces the same RKHS as the shallow ReLU kernel via an endpoint (Puisseux) expansion, so the bottleneck does not shrink the kernel-regime function class while reducing the trainable parameter scaling from dense $O(LN^2)$ to $O(LrN)$. We connect the $I(r) \sim r^{-1/2}$ decay to $1/\sqrt{r}$ sub-Gaussian concentration of the random kernel around the proxy and give a formal bound linking proxy and empirical conditioning (Appendix C.6).

Keywords: Neural Tangent Kernel, Random Features, Low Rank, Depth Scaling, Conditioning

1. Introduction

Low-rank random-feature networks (RF-LR) freeze random feature directions and train only linear readouts (and biases) into a bottleneck of dimension $r \ll N$ at each layer [Zhang et al. \(2025\)](#). In the lazy/NTK regime, training linearizes and reduces to kernel regression with an (approximately) fixed Gram matrix [Jacot et al. \(2018\)](#). This paper asks: *how do depth L and bottleneck dimension r control conditioning-relevant scales for optimization in kernel regression?* We adopt the standard NTK setup (ReLU activation, isotropic Gaussian initialization, sequential infinite-width limit) and analyze the induced kernel at initialization.

Main results and contributions. In the lazy/NTK regime, gradient flow reduces to kernel regression with an (approximately) fixed kernel [Jacot et al. \(2018\)](#). Our main results are:

- **Recursion.** The RF-LR NTK satisfies an explicit infinite-width recursion with an explicit $1/r$ bottleneck factor, yielding a closed-form expansion (Theorem 3 and Corollary 4).
- **Depth/rank scaling and condition number.** For the deterministic proxy mean recursion, depth drives correlations toward 1, the kernel magnitude saturates, and the diagonal–off-diagonal gap decays as $\asymp 1/(rk)$ (Theorem 5). All condition-number statements refer to

the *proxy* Gram matrix. We prove a *lower bound* $\kappa \geq \Omega(r \cdot L)$ in general (Proposition 6). In one special case—equicorrelated data or high-dimensional random data (pairwise correlations $O(1/\sqrt{d})$ w.h.p.)—we give exact conditioning: $\kappa_{\perp} = 1$ or $\kappa_{\perp} = 1 + o(1)$ (Corollary 7). Under a fixed parameter budget, depth and rank trade off from a conditioning perspective.

- **Low rank is enough for the NTK RKHS (three-layer case).** In the three-layer case, the mean RF-LR kernel induces the same RKHS as the shallow ReLU kernel (Corollary 14), i.e. low rank preserves kernel-regime expressivity while making deep kernel models parameter-efficient ($O(rN)$ trainable parameters per layer rather than $O(N^2)$); $L \geq 4$ remains open.

Formal statements: Theorem 3, Theorem 5, Proposition 6, Corollary 7, Corollary 14. Proof sketch: Appendix A.6.

1.1. Architecture

The architecture stacks high-then-low width layers with a bottleneck $r \ll n_1$; we train readouts and biases while keeping feature directions frozen. In the kernel regime, the Gram matrix spectrum controls optimization rates Jacot et al. (2018); Cao et al. (2020). We characterize limiting kernels and spectra under a composition-of-GPs view in the sequential infinite-width limit.

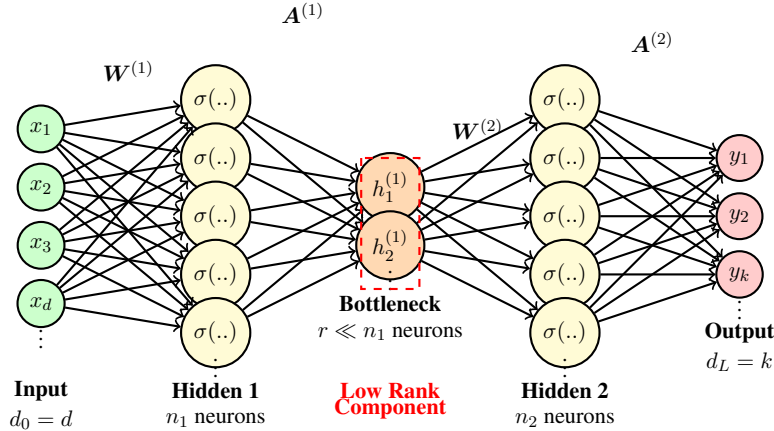


Figure 1: RF-LR: low-rank bottleneck (red dashed) reduces width from n_1 to r ; readouts trained, feature directions frozen.

1.2. Related work

The NTK framework Jacot et al. (2018) characterizes infinitely wide networks as kernel methods. Extensions cover finite-width corrections Arora et al. (2019); Dyer and Gur-Ari (2019), depth scaling Yang (2019); Terjék (2022), and random-matrix limits for Gram spectra Benigni and Paquette (2025). Low-rank architectures reduce parameters and change optimization dynamics Sainath et al. (2013); Novikov et al. (2015); Arora et al. (2018); freezing features isolates r as the key control parameter. LoRA theory situates low-rank fine-tuning between kernel and feature-learning regimes Dayi and Chen (2024); Jang et al. (2024). For ReLU kernels, Bietti and Bach Bietti and

Bach (2021) established the “deep equals shallow” RKHS phenomenon via endpoint expansions; we combine Fisher–Kibble decoupling Fisher (1915); Hotelling (1953) with a Puiseux analysis near $\rho = 1$.

Scope, regime, roadmap. We study the RF-LR NTK in the sequential infinite-width limit and its deterministic proxy; we do not analyze feature learning, adaptive optimizers, or continual learning. Regime: fix n , take $N \rightarrow \infty$, then study concentration as $r \rightarrow \infty$; we do not pursue $n \rightarrow \infty$ bulk spectral laws here. Sections 2–5 define the architecture, derive the recursion, state depth–rank scaling, and give the RKHS identification.

2. Network definition and EOC parameterization

2.1. RF-LR architecture

Let $\mathbf{h}^{(0)}(x) = x \in \mathbb{R}^{d_0}$. For $\ell = 1, \dots, L$,

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)\right) + c^{(\ell)}. \quad (1)$$

Training policy: The readouts $A^{(\ell)}$ and layer biases $c^{(\ell)} \in \mathbb{R}^{d_\ell}$ are trained; the feature directions $w^{(\ell)}$ are frozen random draws (i.i.d. Gaussian). The $1/\sqrt{n_\ell}$ scaling (NTK parameterization) ensures a well-defined infinite-width limit.

Width vs bottleneck dimension. We distinguish the feature width n_ℓ (number of random features) from the output dimension $d_\ell = \dim(\mathbf{h}^{(\ell)})$. The trainable matrix $A^{(\ell)} \in \mathbb{R}^{d_\ell \times n_\ell}$ has columns $A_j^{(\ell)} \in \mathbb{R}^{d_\ell}$. A *bottleneck* layer corresponds to $d_\ell = r \ll n_\ell$, yielding $O(rn_\ell)$ trainable parameters and $\text{rank}(A^{(\ell)}) \leq r$ automatically.

Training parameters, biases, and initialization. Initialization is

$$w_j^{(\ell)} \sim \mathcal{N}(0, I_{d_{\ell-1}}/d_{\ell-1}), \quad A_{ij}^{(\ell)} \sim \mathcal{N}(0, 2), \quad c^{(\ell)} = 0,$$

We use the *edge-of-chaos* (EOC) scaling for the weights so that $\text{Cov}(w_j^{(\ell)}) = I_{d_{\ell-1}}/d_{\ell-1}$ Hayou et al. (2018). In our bottleneck regime we take $d_{\ell-1} = r$ for $\ell \geq 2$, so $w_j^{(\ell)} \in \mathbb{R}^r$ with covariance I_r/r .

We include a layer bias $c^{(\ell)}$ for completeness, but we avoid introducing extra scaling parameters for it. Bias scaling can be absorbed into the learning rate; the infinite-width and NTK structures remain qualitatively identical, and the kernel-regime predictions depend on kernel geometry, not absolute scales.

With this setup in place, we record the standing assumptions and conventions used throughout the paper.

2.2. Assumptions

We collect the standing assumptions and conventions used throughout the paper; expanded discussion is in Appendix A.

- **Sequential infinite width.** We take $n_1 \rightarrow \infty, \dots, n_L \rightarrow \infty$ with i.i.d. Gaussian random features, frozen after initialization, under NTK scaling $1/\sqrt{n_\ell}$ Jacot et al. (2018) (Appendix A.1).
- **Bottleneck regime and scaling.** For $\ell \geq 2$, $d_\ell = r \ll n_\ell$, and we use edge-of-chaos scaling $\text{Cov}(w_j^{(\ell)}) = I_r/r$ (Appendix A.2).
- **Input normalization.** For RKHS statements we focus on normalized inputs (typically $x \in \mathbb{S}^{d-1}$), so kernels are zonal functions of the cosine similarity ρ ; concentration statements keep explicit $\|x\|\|x'\|$ factors when needed.

Homogeneous activation. We work under the positive 1-homogeneity assumption (satisfied by σ); see Assumption 1 in Appendix A. This condition is used for Fisher–Kibble decoupling (Lemma 9) and for separating radial and angular fluctuations in the three-layer kernel.

2.3. Parameterization, signal propagation, and kernels

We briefly record the scaling conventions used in the kernel calculations; details are in Appendix A.2 and Appendix B.1. *EOC scaling:* we choose the variance σ_A so that the forward variance remains stable across depth (edge of chaos), leading to a recursion for the layerwise variance $q^{(\ell)}$ and a corresponding condition on σ_A . *Bottleneck scaling and the $1/r$ factor:* in the bottleneck regime $d_{\ell-1} = r$ for $\ell \geq 2$, the EOC scaling uses $\text{Cov}(w) = I_r/r$, which keeps base and derivative kernels $O(1)$, and the RF-LR NTK recursion carries an explicit prefactor $1/r$ at bottleneck layers. With an abuse of notation, we denote by $\Sigma^{(\ell)}$ the base kernel and by $\dot{\Sigma}^{(\ell)}$ the derivative kernel.

Definition 1 (Base and derivative kernels) *In the sequential infinite-width limit, the layer- ℓ base kernel is*

$$\Sigma^{(\ell)}(x, x') = \mathbb{E}_w \left[\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x')) \right], \quad (2)$$

and the derivative kernel is

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right]. \quad (3)$$

For $\ell = 1$ and centered Gaussian w , $\Sigma^{(1)}$ is rotation-invariant with standard closed forms in terms of input cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$:

$$\Sigma^{(1)}(x, x') = \frac{\|x\| \|x'\|}{2\pi} ((\pi - \theta) \cos \theta + \sin \theta), \quad \theta = \arccos(\rho).$$

We emphasize that these are not the NNGP kernel and the classical derivative kernel appearing in the NTK formulation. Rather, they are the base kernel and the derivative kernel of the Gaussian process $\mathbf{h}^{(\ell)}$ conditional on $\mathbf{h}^{(\ell-1)}$. By rotation invariance of Gaussian weights, these kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$. We will therefore write $\Sigma^{(\ell)}(\rho)$ and $\dot{\Sigma}^{(\ell)}(\rho)$. When no confusion arises (e.g., for $\ell = 1$), we will also abuse notation and write $\sigma(\rho)$ for the scalar kernel function; this σ denotes the kernel-as-a-function-of- ρ , not the activation σ . We can now state the base and derivative kernels precisely.

Given these definitions, we can now state the following composition result.

Theorem 2 (Gaussian process composition) *In the sequential infinite-width limit, the hidden states $\mathbf{h}^{(\ell)}$ form a composition of Gaussian processes:*

- *Conditionally on $\mathbf{h}^{(\ell-1)}$, for any finite collection of inputs $\{x_1, \dots, x_m\}$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian.*
- *The conditional limiting Gaussian has mean zero and covariance matrix $[\Sigma^{(\ell)}(x_i, x_j)]_{i,j=1}^m$, where $\Sigma^{(\ell)}$ is the base kernel defined in Definition 1 and depends on $\mathbf{h}^{(\ell-1)}$*

Proof See Appendix B.2. ■

3. NTK recursion

We now state the main result characterizing the NTK recursion for MMNNs. This recursion governs how the kernel evolves through layers and determines the training dynamics in the infinite-width limit.

Theorem 3 (Infinite-width NTK recursion) *Under NTK parameterization $1/\sqrt{n_\ell}$ and a sequential limit $n_1 \rightarrow \infty, \dots, n_L \rightarrow \infty$ with $\mathbf{w}^{(\ell)} \sim \mathcal{N}(0, \mathbf{I}_{\mathbf{d}_{\ell-1}}/\mathbf{d}_{\ell-1})$ i.i.d. frozen after initialization, the layer- ℓ NTK satisfies*

$$\Theta^{(0)}(x, x') = 0, \tag{4}$$

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \tag{5}$$

$$\Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \Sigma^{(\ell)}(x, x'), \quad \ell \geq 2, \tag{6}$$

where the factor $1/r$ appears for bottleneck layers $\ell \geq 2$ (low-rank regime). For $\ell = 1$, $\Sigma^{(1)}$ is deterministic; for $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields whose fluctuations shrink with the rank r of bottlenecks. The additive constant 1 term comes from the trained layer biases $c^{(\ell)}$ and corresponds to the constant mode (which vanishes after centering).

Furthermore, for a finite-width network initialized at random parameters θ_0 , the empirical NTK $\Theta_{\theta_0}^{(L)}$ is a random kernel (a Gram operator of gradients). Consequently the kernel gradient flow trajectory $t \mapsto f_t$ is also a random object (measurable with respect to θ_0) and satisfies

$$\frac{d}{dt} f_t(x) = - \int \Theta_{\theta_0}^{(L)}(x, x') (f_t(x') - y(x')) d\mu(x'),$$

where μ is the data distribution and y is the target function. In the sequential infinite-width NTK limit, $\Theta_{\theta_0}^{(L)}$ concentrates and converges (in probability) to the deterministic kernel $\Theta^{(L)}$, yielding the corresponding deterministic limiting gradient flow.

Proof: See Appendix B.1, Proof B.1.

We can now derive an explicit closed-form expression for the NTK at any depth by expanding the recursion.

Corollary 4 (General L-layer NTK; closed form) *Under the assumptions of Theorem 3, for any $L \geq 1$, the NTK admits the explicit form:*

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \frac{1}{r^{L-\ell}} \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \frac{1}{r^{\max(1, L-\ell)}} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (7)$$

where the first sum has bias contributions (empty product = 1 when $\ell = L$) and the second has fresh-basis contributions; each bottleneck layer contributes a factor $1/r$. For $L = 2$: $\Theta^{(2)} = 1 + \frac{1}{r}(\dot{\Sigma}^{(2)} + \Sigma^{(1)}\dot{\Sigma}^{(2)} + \Sigma^{(2)})$: the leading 1 is the bias (constant mode); $\frac{1}{r}\dot{\Sigma}^{(2)}$ is the bias-path term (derivative at layer 2 only); $\frac{1}{r}\Sigma^{(1)}\dot{\Sigma}^{(2)}$ and $\frac{1}{r}\Sigma^{(2)}$ are the fresh-basis terms (layers 1 and 2). This matches Theorem 3 by substituting $\Theta^{(1)} = 1 + \Sigma^{(1)}$ into the recursion for $\ell = 2$.

Proof Expanding the recursion in Theorem 3 yields the two sums in (7). A complete proof by induction (including an explicit $L = 3$ expansion) is given in Appendix B.4; see also Appendix 16. ■

Finite-width effects. Beyond the infinite-width limit, one can in principle study finite-width corrections to the kernel via cumulant/resolvent expansions or diagrammatic methods; see Dyer and Gur-Ari (2019) for a Feynman-diagram approach in wide-network asymptotics. These corrections are deferred to future work.

Relation to fully trained MLPs at the edge of chaos. For comparison with the fully trained (full-width) MLP NTK at EOC, see Appendix A.3 and the reference closed form in Proposition 17.

3.1. Large-depth regime: probabilistic kernel and exponential decay

The mean (probabilistic) recursion and the resulting nested conditional expectations are essential for understanding the deep ($L \rightarrow \infty$) behavior, but the full discussion is lengthy. We therefore defer the probabilistic recursion, its nested-expectation form, and the exponential depth-suppression bound to Appendix A.4 (see also Appendix C.5 for related open problems).

The three-layer case admits a compact representation that makes the Fisher–Kibble structure explicit; we present it in Section 5.1.

4. Depth scaling of the RF-LR NTK

4.1. Main results

Random correlations and deterministic proxy. At each layer $\ell \geq 2$, the correlation $\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ is a *random variable*: given the population correlation from the previous layer, Fisher’s law and Lemma 21 give $\mathbb{E}[(\rho_\ell - \varrho(\rho_{\ell-1}))^2] \leq C/r$ and $\mathbb{P}(|\rho_\ell - \varrho(\rho_{\ell-1})| \geq t) \leq 6 \exp(-c r t^2)$, so fluctuations are $O(1/\sqrt{r})$ sub-Gaussian. The NTK recursion passes through this chain of random correlations. The *deterministic proxy* replaces ρ_ℓ by the deterministic iterates $\rho_k = \varrho^{(k-1)}(\rho_1)$ and analyzes the scalar recursion (83); for large r the random kernel concentrates around the proxy (Appendix A.4); a formal bound $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = o_P(1)$ as $r \rightarrow \infty$ and its

implication for empirical conditioning are given in Appendix C.6. In the equicorrelated case, a full proof yields $\|(\hat{K} - K_{\text{proxy}})|_{1^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ (Theorem 25). All condition-number statements in this section refer to the *proxy* Gram matrix; we give a lower bound on its condition number and, in one special case, exact conditioning with high probability.

Depth–rank scaling. The recursion contains an explicit $1/r$ bottleneck factor, which leads to saturation of the kernel magnitude as depth grows, while the diagonal–off-diagonal gap decays with depth in the proxy recursion. Theorem 5 analyzes this deterministic proxy, which approximates the mean path of the random recursion for large r . We refer to Appendix D.4 for notation and kernel expansions.

Theorem 5 (Depth scaling for the deterministic proxy RF-LR NTK) *Fix $r > 1$ and $\rho_1 \in (-1, 1)$. Let $\Theta^{(k)}(\rho_1)$ be the deterministic proxy defined by (83), with $\rho_k = \varrho^{\circ(k-1)}(\rho_1)$ deterministic. Then:*

- **(Correlation alignment)** *The correlation recursion aligns polynomially:*

$$1 - \rho_k = \Theta(k^{-2}).$$

- **(Kernel saturation)** $\Theta^{(k)}(\rho_1) \rightarrow \Theta_\star(r)$ as $k \rightarrow \infty$, where $\Theta_\star(r)$ is the fixed point of the limiting recursion obtained by setting $\dot{s} = s = 1/2$, namely

$$\Theta_\star(r) = \frac{2r + 1}{2r - 1}.$$

Moreover,

$$\Theta^{(k)}(\rho_1) - \Theta_\star(r) = O(k^{-1}).$$

- **(Depth-induced gap decay)** Let $\Theta_{\text{diag}}^{(k)} := \Theta^{(k)}(1)$ (on-diagonal value, i.e. identical inputs) and $\Theta_{\text{off}}^{(k)} := \Theta^{(k)}(\rho_1)$ (off-diagonal value for an input pair with initial cosine similarity ρ_1). Then

$$\Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)} = \Theta\left(\frac{1}{r^k}\right),$$

for large k .

Proof See Appendix D.7. ■

Proposition 6 (Lower bound on condition number for the proxy) *In the setting of Theorem 5, consider the deterministic proxy kernel $\Theta^{(L)}(\rho)$ and form the $n \times n$ Gram matrix $\mathbf{M}$ with entries $M_{ij} = \Theta^{(L)}(\rho_{ij})$ for pairwise input cosine similarities $\rho_{ij} = \langle x_i, x_j \rangle / (\|x_i\| \|x_j\|) \in (-1, 1)$. Let $\lambda_{\max}$ and $\lambda_{\min}$ denote the maximum and minimum eigenvalues of $\mathbf{M}$ restricted to $\mathbf{1}^\perp$. Assume: (i) $n \geq 3$; (ii) there exists a pair (i, j) with ρ_{ij} bounded away from 1; (iii) the dataset is not equicorrelated (the ρ_{ij} for $i \neq j$ are not all equal). Then $\lambda_{\max} = \Theta(1)$, $\lambda_{\min} \leq O(1/(rL))$, and the condition number $\kappa = \lambda_{\max}/\lambda_{\min}$ satisfies $\kappa \geq \Omega(r \cdot L)$. When assumption (iii) is violated (equicorrelated data), the proposition does not apply; instead Corollary 7 gives the best-case $\kappa_\perp = 1$ (and similarly for approximately equicorrelated high-dimensional data).*

Proof See Appendix D.5. ■

Theorem 7 (Operator norm on $\mathbf{1}^\perp$ in the equicorrelated case) Fix $n, L \geq 1$, $r \geq 2$, and the equicorrelated setup $\rho_{1,ii} = 1$, $\rho_{1,ij} = \rho_0$ for $i \neq j$ with $\rho_0 \in (-1, 1)$. Let $\hat{K}$ be the $n \times n$ empirical RF-LR NTK Gram matrix and K_{proxy} the deterministic proxy with entries $(K_{\text{proxy}})_{ij} = \Theta^{(L)}(\rho_{1,ij})$ (Definition D.4). Then there exist constants $C, c > 0$ (depending on L, ρ_0) such that for any $\epsilon > 0$,

$$\mathbb{P}\left(\left\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\right\|_{\text{op}} \geq \epsilon\right) \leq C \exp\left(-c \frac{r \epsilon^2}{L^2}\right) + C \exp\left(-c \frac{r \epsilon}{L}\right).$$

In particular, $\left\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\right\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ as $r \rightarrow \infty$ with n, L fixed. Hence if $r = O(L^2)$, then $\left\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\right\|_{\text{op}} = O_P(L/r)$.

Proof See Appendix C.6. ■

Corollary 8 (Exact conditioning for equicorrelated and high-dimensional random data) In the setting of Proposition 6, the following hold.

- **Equicorrelated data.** If $\rho_{ij} = \rho_0$ for all $i \neq j$ (and $\rho_{ii} = 1$), then on $\mathbf{1}^\perp$ all eigenvalues of $\mathbf{M}$ equal $\lambda_\perp = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0) = \Theta(1/(rL))$, so the condition number on the mean-zero subspace is $\kappa_\perp = 1$.
- **High-dimensional random data.** If the inputs $x_1, \dots, x_n \in \mathbb{S}^{d-1}$ are drawn i.i.d. uniformly from the unit sphere with $d \rightarrow \infty$, then with high probability the pairwise cosine similarities satisfy $\max_{i \neq j} |\rho_{ij}| = O(1/\sqrt{d})$, so the data are approximately equicorrelated. In that regime, the proxy Gram matrix has, with high probability, all eigenvalues on $\mathbf{1}^\perp$ equal to $\lambda_\perp = \Theta(1)(1 + o(1))$, hence $\kappa_\perp = 1 + o(1)$. On that high-probability event the same operator-norm bound on $\mathbf{1}^\perp$ as in the equicorrelated case applies (Theorem 25 with $\rho_0 = O(1/\sqrt{d})$).

Proof See Appendix D.6. ■

Limited scope of exact conditioning. Equicorrelated and high-dimensional i.i.d. spherical data are special; the general conditioning guarantee in this section is the proxy lower bound $\kappa \geq \Omega(r \cdot L)$ (Proposition 6).

Remark 9 (Proxy vs. optimization) Proposition 6 gives a lower bound on the condition number of the proxy Gram matrix; we do not claim that this bound governs optimization in the finite-width network. For the proxy and non-equicorrelated data, depth drives the kernel toward near-constant entries (gap $\Theta(1/(rL))$) and $\kappa \geq \Omega(rL)$. The actual network uses random correlations with $O(1/\sqrt{r})$ sub-Gaussian fluctuations (Appendix A.4), so for large r the random kernel concentrates around the proxy. The condition number is relevant for kernel regression (stability and convergence speed depend on $\lambda_{\max}/\lambda_{\min}$); under a fixed budget $O(LrN)$, depth and rank trade off from a conditioning perspective, and the choice of r and L is a matter of approximation power (effective depth and RKHS expressivity).

5. Low rank is enough for the NTK RKHS (three-layer mean kernel)

We show that low rank does not shrink the kernel-regime function class: the mean three-layer RFLR kernel induces the same RKHS as the shallow ReLU kernel.

Roadmap. We first derive the Fisher–Kibble decoupling (Section 5.1), which separates angular (correlation) and radial (norm) randomness and yields a compact three-layer empirical NTK form. We then obtain a Puiseux expansion of the mean kernel near $\rho = 1$ (Section 5.2), which controls the RKHS. The leading exponent matches the shallow ReLU kernel, so the RKHSs coincide (Corollary 14). The analysis is restricted to three layers because the Fisher–Kibble integral admits a closed form for a single bottleneck; for $L \geq 4$ (depth at least four layers), the nested Fisher-chain expectations require a Laplace-type expansion that remains open (Appendix C.5).

5.1. Microscopic distributions: Fisher–Kibble decoupling

We can now state a compact representation under a homogeneity assumption.

Lemma 10 (Fisher–Kibble decoupling) *Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#); [Hotelling \(1953\)](#), centered at the true correlation ρ .
- (u, v) follow Kibble’s [Kibble \(1941\)](#) bivariate Gamma (chi-square) law.
- Angular and radial parts are independent: $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Explicit densities. Fisher’s density (for $r > 2$) is

$$p_{\text{Fisher}}(\rho_1) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-\rho_1^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho\rho_1)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho\rho_1}{2}\right).$$

Kibble’s bivariate chi-square density for $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ is

$$p_{\text{Kibble}}(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} \exp\left(-\frac{u+v}{2(1-\rho^2)}\right)}{\Gamma(r/2) (2(1-\rho^2))^{\frac{r}{2}+1} \rho^{\frac{r/2-1}{2}}} I_{\frac{r}{2}-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right), \quad u, v \geq 0,$$

where I_ν is the modified Bessel function of the first kind.

Proof: See Appendix C.2, Proof C.2.

Homogeneity assumption enabling decoupling. By positive 1-homogeneity and rotational invariance, the base kernel factorizes and $(\rho_1, \|x_1\|, \|y_1\|)$ decouple into angular (Fisher) and radial (Kibble) components; see Appendix 20 and Appendix C.2 for details, explicit densities, and the induced compact three-layer empirical NTK form.

Compact three-layer empirical NTK form. Under the assumptions of Theorem 3 with $L = 2$, the three-layer empirical RF-LR NTK admits the representation

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x'), \quad \Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \quad (8)$$

and, under the positive 1-homogeneity assumption (satisfied by σ), the layer-2 fields can be written in terms of the Fisher angular variable ρ_1 and the Kibble radial factor $w_r := \sqrt{uv}/r$ as

$$\dot{\Sigma}^{(2)}(x, x') = 1 - \frac{\arccos(\rho_1)}{\pi}, \quad \Sigma^{(2)}(x, x') = w_r \Sigma^{(1)}(\rho_1). \quad (9)$$

Combining (8)–(9) yields an explicit three-layer kernel in the decoupled microscopic variables (ρ_1, u, v) ; equivalently,

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \left(1 - \frac{\arccos(\rho_1)}{\pi}\right) + \frac{1}{r} w_r \Sigma^{(1)}(\rho_1). \quad (10)$$

See Remark 20 (Appendix C.2) for the full derivation, the explicit Fisher and Kibble densities, and the induced compact form (10).

5.2. Endpoint expansion via hypergeometric analysis

The RKHS of zonal kernels on the sphere is controlled by the Puiseux behavior near $\rho = \pm 1$ Bietti and Bach (2021). For the mean three-layer kernel, the key step is to analyze the Fisher expectation $\mathbb{E}[\arccos(\hat{\rho}_r)]$ as $\rho \rightarrow 1$.

Proposition 11 (Puiseux expansion of $\mathbb{E}(\arccos(\hat{\rho}_r))$ near $\rho = 1$) *Let $r > 2$ and $\hat{\rho}_r \sim \text{Fisher}(\rho, r)$. Write $\rho = 1 - t$ with $t \downarrow 0$. Then*

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}),$$

where $I(r) \in (0, \infty)$ admits the closed form

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2}-1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r-\frac{1}{2}) \Gamma(r-\frac{3}{2})}{\Gamma(r-1)^2}. \quad (11)$$

Proof [Proof sketch] Insert Fisher’s density (Appendix 20) into $\int_{-1}^1 \arccos(s) p_{\text{Fisher}}(s \mid 1-t, r) ds$ and change variables $s = 1 - v$. Near $t, v \rightarrow 0$, use $\arccos(1-v) = \sqrt{2v} + O(v^{3/2})$ and apply the hypergeometric connection formula (DLMF §15.8.2) to replace ${}_2F_1(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; 1 - \frac{t+v}{2})$ by the constant $C_1(r)$ up to $O((t+v)^{r-3/2})$. Then the leading integral reduces, after $w = v/t$, to a Beta integral and yields (11). See Appendix C.9 for the full proof. ■

Proposition 12 ($I(r)$ decays as $1/\sqrt{r}$) *Let $I(r)$ be defined by (11). Then $I(r) = O(1/\sqrt{r})$ as $r \rightarrow \infty$.*

Proof [Proof sketch] Apply Stirling’s formula to the Gamma ratios in (11). A detailed proof is given in Proposition 32 (Appendix C.10). ■

Corollary 13 (Puiseux expansion of the mean three-layer NTK near $\rho = 1$) *Let $\tilde{\Theta}^{(2)}(\rho)$ denote the mean three-layer RF-LR NTK and let $K_\infty(\rho)$ denote the deterministic $r \rightarrow \infty$ limit kernel,*

$$K_\infty(\rho) = \Theta^{(1)}(\rho) \left(1 - \frac{\arccos(\rho)}{\pi} \right) + \Sigma^{(1)}(\rho) + 1,$$

as in Appendix C.9. Writing $\rho = 1 - t$ with $t \downarrow 0$, one has

$$\tilde{\Theta}^{(2)}(1 - t) = 1 + \frac{1}{r} (K_\infty(1) - 1) - \frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}), \quad (12)$$

where $I(r)$ is as in Proposition 10. In particular, since $I(r) = O(1/\sqrt{r})$ (Proposition 11), the bracket equals $2/\pi + O(1/\sqrt{r})$, so the $t^{1/2}$ coefficient is $\frac{2}{\pi r} + O(r^{-3/2})$.

Proof [Proof sketch] Combine the three-layer mean-kernel identity (Appendix C.9) with Proposition 10 and the ReLU endpoint expansion $\arccos(1 - t) = \sqrt{2t} + O(t^{3/2})$. A complete proof is given in Appendix C.9. ■

5.3. RKHS equivalence via endpoint behavior

We now state the RKHS characterization result used to translate Puiseux endpoint behavior into an RKHS statement.

Theorem 14 (Bietti–Bach RKHS criterion Bietti and Bach (2021), Theorem 1) *Let K_1, K_2 be zonal kernels on $\mathbb{S}^{d-1}$ that are C^∞ on $(-1, 1)$. Suppose that as $t \downarrow 0$,*

$$K_i(1 - t) = K_i(1) - c_i t^\alpha + O(t^{\alpha+\eta}), \quad c_i > 0,$$

for the same exponent $\alpha \in (0, 1)$ and some $\eta > 0$ (and likewise at $\rho = -1$ if needed). Then K_1 and K_2 induce the same RKHS on $\mathbb{S}^{d-1}$, with equivalent norms. In Bietti and Bach (2021), Theorem 1 states that the eigenvalue decay of the associated integral operator (hence the RKHS) is determined by the differentiability of the kernel at the endpoints; matching Puiseux exponents α yields the same decay and thus the same RKHS. Equivalently, the Mercer eigenvalues of the kernel integral operator on the sphere decay with a rate determined by α ; matching α therefore yields the same RKHS (same eigenvalue decay and equivalent norms).

Proof Direct consequence of (Bietti and Bach, 2021, Theorem 1). ■

Corollary 15 (Low rank is enough for the NTK RKHS: three-layer mean kernel) *Under the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the three-layer mean NTK $\tilde{\Theta}^{(2)}$ induces the same RKHS as the shallow ReLU kernel. In particular, the corresponding RKHSs coincide as sets with equivalent norms.*

Proof [Proof sketch] The mean three-layer kernel is zonal. Proposition 10 gives a Puiseux expansion near $\rho = 1$ with leading exponent $t^{1/2}$; Proposition 11 shows the r -dependent coefficient is an $O(1)$ perturbation of the shallow ReLU coefficient. Since the shallow ReLU kernel has the same endpoint exponent $1/2$, Theorem 13 yields RKHS equivalence. See Appendix C.9 for a detailed argument. ■

Open problem: depth $L \geq 4$. Extending RKHS equivalence to general depth (at least four layers) requires controlling endpoint expansions through the full Fisher chain; we leave this to future work (Appendix C.9).

Concentration (appendix). We also prove that the empirical three-layer NTK concentrates around its deterministic limit with sub-Gaussian tails in r ; see Corollary 28 and Appendix C.8. The $I(r) = O(1/\sqrt{r})$ scaling (Proposition 32) is consistent with the fact that Fisher–Kibble corrections to the mean kernel’s leading endpoint coefficient vanish as r grows.

6. Conclusion

This work develops an NTK analysis for low-rank random-feature architectures (RF-LR/MMNN) under explicit assumptions. The paper contains no experiments; empirical validation is left for future work.

Main: depth/rank scaling and condition number. The trainable parameter budget scales as $O(LrN)$. In the deterministic proxy mean recursion, depth drives correlations toward 1, the kernel magnitude saturates, and the diagonal–off-diagonal gap decays as $\asymp 1/(rk)$ (Theorem 5). We prove a lower bound on the condition number of the *proxy* Gram matrix, $\kappa \geq \Omega(r \cdot L)$ (Proposition 6), and one result giving exact conditioning for equicorrelated and high-dimensional random data (Corollary 7). Under a fixed budget, depth and rank trade off from a conditioning perspective; the choice of r and L is a matter of approximation power (e.g. effective depth and RKHS expressivity).

Recursion. We provide an explicit NTK recursion for arbitrary depth in RF-LR and a closed-form expansion (Theorem 3, Corollary 4).

Low rank is enough for the NTK RKHS (three layers). In the NTK regime, the three-layer RF-LR mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 14), showing that the bottleneck does not reduce the kernel-regime function class while lowering the trainable parameter scaling from dense $O(LN^2)$ to $O(LrN)$.

Concentration in the bottleneck dimension. For the three-layer case, the empirical kernel concentrates around its deterministic limit $K_\infty(\rho)$ with tails controlled by the bottleneck dimension r (Theorem 27, Corollary 28). For equicorrelated data, a full proof in the appendix gives $\|(\hat{K} - K_{\text{proxy}})|_{\mathbf{1}^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ (Theorem 25), linking proxy and empirical conditioning on the mean-zero subspace.

Condition number (kernel GD). For squared loss in the lazy/NTK regime with mean-zero targets, stability and convergence speed depend on the condition number $\lambda_{\max}/\lambda_{\min}$ of the centered Gram matrix: stability requires $\eta < 2/\lambda_{\max}$, and the best fixed-step linear rate depends on $\lambda_{\max}/\lambda_{\min}$. Our results give a lower bound on this quantity for the proxy and, in one setting, its exact value with high probability. We do not claim to govern optimization; the link between proxy condition number and GD convergence or sample complexity is not made precise in this work.

7. Discussion

This work leaves several natural directions open. We emphasize that our current scope is the lazy/NTK regime under explicit standard assumptions; extending beyond this setting requires additional ideas.

A recurring qualitative message, shared with depth analyses for fully trained MLPs at the edge of chaos [Terjék and González-Sánchez \(2025\)](#); [Hayou et al. \(2022\)](#), is that increasing depth tends to drive correlations toward 1, so layerwise kernels become nearly constant and the deep mean kernel approaches a rank-one structure. In RF-LR, the mean-kernel infinite-depth limit collapses after centering unless one rescales time/learning rate (Remark 38 in Appendix D.9).

A technical feature of our setup is that freezing the random features removes some cross-layer dependencies that appear in fully trained networks. Combined with Fisher–Kibble decoupling [Fisher \(1915\)](#); [Hotelling \(1953\)](#), this makes certain calculations more explicit and suggests a path toward extending the analysis to joint large- n regimes using resolvent replacement and Gaussian-equivalence methods developed for shallow two-layer NTKs [Benigni and Paquette \(2025\)](#).

The recursion (Theorem 3) suggests organizing finite-width corrections as an expansion in the bottleneck dimension r . Making this program rigorous for general depth would require controlling the growth of cross-terms produced by the recursion and propagating concentration through the composed Gaussian-process layers.

On the theoretical front, several important questions remain open and constitute an immediate research agenda. Extending the exponential concentration bounds to general depth L layers requires controlling nested conditional means through L layers and managing a quadratic number $O(L^2)$ of cross-terms arising from the recursive NTK structure. The technical challenge lies in tracking how the random correlations ρ_ℓ and norm products $\|h^{(\ell)}(x)\| \|h^{(\ell)}(x')\|$ propagate through layers while maintaining exponential concentration rates. A **non-Gaussian process propagation analysis** distinct from NNKP fits would provide complementary insights beyond the kernel regime, potentially revealing how feature learning emerges from deviations from Gaussianity. An important future direction is to establish **bulk spectral laws** for the empirical RF-LR NTK Gram matrix in a quadratic scaling regime $n \rightarrow \infty$, adapting the Benigni–Paquette framework beyond shallow two-layer kernels to the present low-rank random-feature structure and to depth $L \geq 2$ [Benigni and Paquette \(2025\)](#).

Depth: what extends to general L vs. what remains open. All **conditioning** results in this paper hold for *arbitrary depth* L : proxy depth scaling, condition-number lower bound $\kappa \geq \Omega(r \cdot L)$, exact conditioning for equicorrelated and high-dimensional data, and the rigorous proxy–empirical bound on $1^\perp$ (Theorem 25) are proved for general L . In contrast, **RKHS equivalence** (“low rank is enough”) is established only for the three-layer mean kernel; extending it to $L \geq 4$ is open. The three-layer proof relies on a one-dimensional Fisher expectation and a ${}_2F_1$ connection formula for the Puiseux expansion near $\rho = 1$. For $L \geq 4$, the mean kernel is an $(L - 1)$ -fold expectation over a Fisher-type chain, and the conditional expectations involved differ from the unconditional mean kernel; a full proof would require controlling backward chain densities and nested hypergeometric/Laplace integrals. Possible directions include an inductive argument that the leading Puiseux exponent stays $1/2$ at each layer or a characterization that avoids full chain marginals. See Appendix ?? for details.

Additional theoretical extensions hold particular promise. The current analysis focuses on the three-layer case for tractability where the RKHS is concerned; extending the RKHS identification to arbitrary depth L while maintaining explicit formulas remains an open challenge. Investigating how the edge-of-chaos (EOC) parameterization affects finite-width corrections and the Nr scaling mentioned in the architecture definition would provide concrete quantitative bounds. Understanding how the curse of dimensionality impacts normalization choices (unit norm weights vs. biases)

depending on data geometry (hypercube vs. spherical) would provide principled guidance for practitioners.

The experimental validation of our theoretical predictions constitutes an equally important and concrete next step. We envision a comprehensive program that includes **confirming lazy-kernel predictions** and precisely mapping the boundary of lazy training in practice; conducting systematic **three-layer width sweeps** to test the $O(1/n) + O(1/r)$ finite-width corrections suggested by the expansion and identify minimal (n, r) configurations where kernel predictions become accurate; and a detailed **comparison suite with MLPs** that probes centered conditioning and optimization time scales at matched parameter counts. These studies would verify the **exponential concentration** rates (Theorem 27) and demonstrate the **practical accessibility of the kernel regime** for low-rank random features.

Finally, it would be valuable to connect the low-rank random-feature recursion studied here with broader frameworks for infinite-width limits (e.g. tensor programs Yang (2020)) and with depth-dependent NTK spectral analyses for MLPs at the edge of chaos Terjék and González-Sánchez (2025), while maintaining explicit control of the bottleneck scaling.

Our present scope is deliberately focused on the **lazy/NTK regime**, where analytical tractability is greatest and the advantages of low-rank structure are most clearly visible. Completing finite-width formulas, spectrum characterizations, and experimental validations would strengthen the connection between the asymptotic theory and practical regimes.

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Appendix A. Notation and standing assumptions

A.1. Standing assumptions (expanded)

We summarize the conventions used throughout the main text.

- **Sequential infinite width.** Widths $n_1, \dots, n_L$ tend to infinity sequentially under NTK scaling $1/\sqrt{n_\ell}$. Random features $w^{(\ell)}$ are i.i.d. Gaussian and frozen after initialization.
- **Training policy.** Only the readouts $A^{(\ell)}$ (and layer biases $c^{(\ell)}$) are trained; all other parameters are frozen.
- **Bottleneck regime.** For $\ell \geq 2$, the representation dimension is $d_\ell = r \ll n_\ell$. This induces an explicit $1/r$ factor in the RF-LR NTK recursion at bottleneck layers.
- **Zonal kernels.** By rotational invariance of Gaussian features, base and derivative kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$. For RKHS results we typically take $x \in \mathbb{S}^{d-1}$.
- **Homogeneity.** We use Assumption 1 (positive 1-homogeneity) for Fisher–Kibble decoupling and for separating radial and angular components in three-layer calculations.

Assumption 1 (Homogeneous activation) *The activation function σ is positively 1-homogeneous:*

$$\sigma(\alpha u) = \alpha \sigma(u) \quad \text{for all } \alpha \geq 0, u \in \mathbb{R}.$$

In particular, σ satisfies this property. This assumption is used for Fisher–Kibble decoupling (Lemma 9) and for separating radial and angular fluctuations in the three-layer kernel.

A.2. EOC scaling and the origin of $1/r$

Let $\Sigma_w = \text{Cov}(w_j^{(\ell)})$. Writing $q^{(\ell)} = \mathbb{E} \|h^{(\ell)}(x)\|^2$ for a fixed unit-norm input, weight independence yields the variance recursion

$$q^{(\ell)} = \frac{\sigma_A^2}{2} q^{(\ell-1)} \text{Tr}(\Sigma_w),$$

The edge-of-chaos (EOC) condition chooses σ_A so that the linearized coefficient equals 1,

$$\frac{\sigma_A^2}{2} \text{Tr}(\Sigma_w) = 1,$$

preventing exploding/vanishing signal propagation across depth. In particular, under $\Sigma_w = I_{d_{\ell-1}}/d_{\ell-1}$ one has $\text{Tr}(\Sigma_w) = 1$ and $\sigma_A = \sqrt{2}$.

In the bottleneck regime $d_{\ell-1} = r$ for $\ell \geq 2$, EOC prescribes $\Sigma_w = I_r/r$. For ReLU, the (normalized) derivative kernel has the closed form

$$\bar{\Sigma}^{(\ell)}(\rho) = \mathbb{P}(Z > 0, Z' > 0) = \frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \in \left[0, \frac{1}{2}\right],$$

so $\dot{\Sigma}^{(\ell)}(x, x') = \bar{\Sigma}^{(\ell)}(\rho_\ell)$ remains $O(1)$ under EOC and is uniformly bounded by $1/2$. The RF-LR NTK recursion (Theorem 3) carries an explicit prefactor $1/r$ at bottleneck layers because gradients propagate through an r -dimensional bottleneck and the corresponding Jacobian metric concentrates on an r -dimensional subspace; see Appendix B.1 for the detailed derivation.

A.3. Comparison with the classical fully-trained NTK recursion

For comparison, consider a standard fully-connected MLP where *all* weights (and biases) are trained under NTK parameterization. In the sequential infinite-width limit, the classical NTK recursion takes the form Jacot et al. (2018)

$$\Theta_{\text{MLP}}^{(0)}(x, x') = 0, \quad \Theta_{\text{MLP}}^{(\ell)}(x, x') = \Theta_{\text{MLP}}^{(\ell-1)}(x, x') \dot{K}^{(\ell)}(x, x') + K^{(\ell)}(x, x'), \quad \ell \geq 1, \quad (13)$$

where $K^{(\ell)}$ is the (deterministic) NNGP kernel at layer ℓ and $\dot{K}^{(\ell)}$ the associated derivative kernel (both obtained from the signal-propagation recursion). Unrolling (13) yields one contribution per layer (a sum of L terms, each propagated through subsequent $\dot{K}^{(k)}$ factors).

In contrast, the RF-LR recursion of Theorem 3 (i) carries explicit $1/r$ prefactors at bottleneck layers, and (ii) uses conditional base/derivative kernels $\Sigma^{(\ell)}, \dot{\Sigma}^{(\ell)}$ rather than the classical NNGP kernels $K^{(\ell)}, \dot{K}^{(\ell)}$. These differences lead to the $2L$ -term expansion in Corollary 4, reflecting separate bias-path and fresh-basis contributions, each propagated through subsequent derivative kernels and weighted by bottleneck factors.

Relation to fully trained MLPs at the edge of chaos. For fully trained (full-width) MLPs at the edge of chaos, the limiting NTK admits a single-layer sum structure (one contribution per layer) without RF-LR bottleneck prefactors; see Proposition 17 for a reference closed form Terjék and González-Sánchez (2025). In contrast, the RF-LR expansion (7) contains $2L$ terms because the recursion has both bias-path and fresh-basis-path contributions, each propagated through subsequent derivative kernels and weighted by explicit $1/r$ factors at bottlenecks.

A.4. Probabilistic recursion, GP concatenation, and bottleneck scaling

Random correlation chain and Fisher’s law. At each layer $\ell \geq 2$, the pair $(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ is a $2r$ -dimensional Gaussian conditional on lower-layer randomness. The *layer- ℓ sample correlation*

$$\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x')) = \frac{\langle \mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x') \rangle}{\|\mathbf{h}^{(\ell-1)}(x)\| \|\mathbf{h}^{(\ell-1)}(x')\|}$$

is a *random variable*. Given the population correlation $\rho = \varrho(\rho_{\ell-1})$ (the conditional mean under the ReLU cosine map), Fisher’s law Fisher (1915); Hotelling (1953) and Lemma 21 give $\mathbb{E}[(\rho_\ell - \rho)^2] \leq C/r$ and $\mathbb{P}(|\rho_\ell - \rho| \geq t) \leq 6 \exp(-c r t^2)$. Thus the chain $\rho_1 \rightarrow \rho_2 \rightarrow \dots \rightarrow \rho_L$ is a *Markov chain of random variables* with $O(1/\sqrt{r})$ fluctuations at each transition. The NTK recursion passes through these random correlations successively: $\Sigma^{(\ell)} = \bar{\Sigma}^{(\ell)}(\rho_\ell)$, $\dot{\Sigma}^{(\ell)} = \bar{\dot{\Sigma}}^{(\ell)}(\rho_\ell)$ with random ρ_ℓ . The deterministic proxy (Definition D.4) replaces this random path by the deterministic iterates $\rho_k = \varrho^{\circ(k-1)}(\rho_1)$, which represent the mean/infinite- r limit path.

Kernel randomness and GP concatenation. For $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields because they depend on $\mathbf{h}^{(\ell-1)}$, which is the output of a (conditional) Gaussian process. By Theorem 2, the hidden states form a composition of GPs: each layer’s output is conditionally Gaussian given the previous layer, but the unconditional law is a deep GP. In the bottleneck regime, the pair $(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ is a $2r$ -dimensional Gaussian conditional on lower-layer randomness, with covariance determined by $\Sigma^{(\ell-1)}$. The base kernel $\Sigma^{(\ell)}$ and derivative kernel $\dot{\Sigma}^{(\ell)}$ are thus expectations over the frozen features $\mathbf{w}^{(\ell)}$, conditional on this Gaussian pair, and inherit randomness through $\Sigma^{(\ell-1)}$.

Bottleneck scaling and explicit ReLU bounds. At a bottleneck layer $\ell \geq 2$, the RF-LR NTK recursion carries an explicit prefactor $1/r$ (Theorem 3). For ReLU and isotropic weights with $\text{Cov}(w) = I_r/r$, one has

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right] = \bar{\Sigma}^{(\ell)}(\rho_\ell),$$

where $\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ and $\bar{\Sigma}^{(\ell)}(\rho)$ denotes the corresponding scalar derivative kernel evaluated at cosine similarity $\rho \in [-1, 1]$. Under the EOC normalization $\text{Tr}(\text{Cov}(w)) = 1$, the ReLU derivative kernel has the explicit form

$$\bar{\Sigma}^{(\ell)}(\rho) = \mathbb{P}(Z > 0, Z' > 0) = \frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \in \left[0, \frac{1}{2}\right],$$

where (Z, Z') is centered Gaussian with correlation ρ . In particular, $\sup_{\rho \in [-1, 1]} \bar{\Sigma}^{(\ell)}(\rho) = 1/2$, uniformly in ℓ . Similarly, $\Sigma^{(\ell)}(x, x') = \mathbb{E}_w[\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x'))]$ is an $O(1)$ scalar kernel, and the $1/r$ factors enter through the recursion.

Probabilistic recursion (nested expectations). The mean (probabilistic) NTK $\tilde{\Theta}^{(L)} = \mathbb{E}[\Theta^{(L)}]$ satisfies a recursion involving expectations of products:

$$\tilde{\Theta}^{(1)} = 1 + \mathbb{E}[\Sigma^{(1)}], \quad \tilde{\Theta}^{(\ell)} = 1 + \frac{1}{r} \mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] + \frac{1}{r} \mathbb{E}[\Sigma^{(\ell)}], \quad \ell \geq 2. \quad (14)$$

To emphasize the nested structure, write $\mathbb{E}_{\leq \ell-1}$ for expectation over all randomness up to depth $\ell-1$, and $\mathbb{E}_{w^{(\ell)}}[\cdot | \mathbf{h}^{(\ell-1)}]$ for expectation over the frozen feature direction $w^{(\ell)}$ at layer ℓ , conditional on the previous hidden states. Then the mean kernels are iterated conditional expectations:

$$\mathbb{E}[\Sigma^{(\ell)}(x, x')] = \mathbb{E}_{\leq \ell-1} \left[\mathbb{E}_{w^{(\ell)}} [\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x')) | \mathbf{h}^{(\ell-1)}] \right],$$

$$\mathbb{E}[\Theta^{(\ell-1)}(x, x') \dot{\Sigma}^{(\ell)}(x, x')] = \mathbb{E}_{\leq \ell-1} \left[\Theta^{(\ell-1)}(x, x') \cdot \mathbb{E}_{w^{(\ell)}} [\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 | \mathbf{h}^{(\ell-1)}] \right].$$

Since $\Theta^{(\ell-1)}$ and $\Sigma^{(\ell)}, \dot{\Sigma}^{(\ell)}$ share the randomness of $\mathbf{h}^{(\ell-1)}$, one has $\mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] \neq \tilde{\Theta}^{(\ell-1)} \cdot \mathbb{E}[\dot{\Sigma}^{(\ell)}]$ in general; controlling this dependence (and the resulting nested conditional expectations) is the main technical point for a rigorous deep mean-kernel theory.

Exponential depth suppression (ReLU, bottlenecks). Assume for simplicity that all layers $\ell \geq 2$ are bottlenecks with the same rank r . For ReLU under EOC we have the explicit uniform bound

$$\bar{\Sigma}^{(\ell)}(\rho) \leq c_0 := \frac{1}{2}, \quad \forall \rho \in [-1, 1], \forall \ell \geq 2.$$

From the closed form (7), the contribution from layer ℓ to $\Theta^{(L)}$ is multiplied by $\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}$. Writing $\dot{\Sigma}^{(k)} = \bar{\Sigma}^{(k)}(\rho_k)$, we obtain

$$\frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)} \leq \frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \bar{\Sigma}^{(k)}(\rho_k) \leq \frac{c_0^{L-\ell}}{r^{L-\ell}}.$$

Hence, contributions from early layers ($\ell \ll L$) decay exponentially in $L - \ell$ when $c_0/r < 1$, and the NTK becomes increasingly localized to the top of the network as depth grows.

Proposition 16 (Effective depth decomposition and a $\log r$ window) *Assume the uniform derivative-kernel bound $\dot{\Sigma}^{(\ell)}(x, x') \leq c_0$ for all $\ell \geq 2$ and all input pairs (x, x') , and assume that for a given (x, x') one has a uniform base-kernel bound*

$$|\Sigma^{(\ell)}(x, x')| \leq C_\Sigma(x, x'), \quad \forall \ell \in \{1, \dots, L\}.$$

Fix $L \geq 2$, $r > c_0$, and an integer $m \in \{1, \dots, L\}$. Define the top- m truncation $\Theta_{\text{top},m}^{(L)}(x, x')$ by restricting both sums in the explicit expansion (7) to indices $\ell \in \{L - m + 1, \dots, L\}$. Then

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top},m}^{(L)}(x, x')| \leq \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^m.$$

In particular, taking $m = \lceil \alpha \log r \rceil$ (for any $\alpha > 0$) yields a $\log r$ effective-depth window: the contribution of layers $\ell \leq L - \lceil \alpha \log r \rceil$ is at most

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top}, \lceil \alpha \log r \rceil}^{(L)}(x, x')| \leq \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^{\lceil \alpha \log r \rceil} = o(r^{-p})$$

for every fixed $p > 0$ as $r \rightarrow \infty$.

Proof sketch. The remainder is the sum of terms in (7) with $\ell \leq L - m$; each such term carries a factor $\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)} \leq c_0^{L-\ell}$ and hence $(c_0/r)^{L-\ell}$. Summing over $\ell \leq L - m$ yields a geometric-series tail; substituting $m = \lceil \alpha \log r \rceil$ gives the stated decay.

Proof By definition of $\Theta_{\text{top},m}^{(L)}$, the remainder is the sum of the terms in (7) with $\ell \leq L - m$. For such ℓ , we have $L - \ell \geq m$, and by the derivative-kernel bound $\dot{\Sigma}^{(k)}(x, x') \leq c_0$ we obtain

$$\frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \leq \left(\frac{c_0}{r}\right)^{L-\ell}.$$

Moreover, for $\ell \leq L - m \leq L - 1$ the base-kernel term carries the same prefactor $r^{-(L-\ell)}$ in (7), hence

$$\frac{1}{r^{L-\ell}} |\Sigma^{(\ell)}(x, x')| \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \leq C_\Sigma(x, x') \left(\frac{c_0}{r}\right)^{L-\ell}.$$

Summing these bounds over $\ell \leq L - m$ and using a geometric-series tail gives

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top},m}^{(L)}(x, x')| \leq (1 + C_\Sigma(x, x')) \sum_{j=m}^{\infty} \left(\frac{c_0}{r}\right)^j = \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^m,$$

as claimed. The $\log r$ specialization follows by substituting $m = \lceil \alpha \log r \rceil$ and observing that $(c_0/r)^{\lceil \alpha \log r \rceil} = o(r^{-p})$ for every fixed $p > 0$. $\blacksquare$

	MLP at EOC Terjék and González-Sánchez (2025)	RF-LR (this work)
Trainable	all weight layers	readouts $A^{(\ell)}$, biases $c^{(\ell)}$; frozen $w^{(\ell)}$
Scaling	depth L , Δ_ϕ	depth L , bottleneck r
Kernel magnitude	uncentered scale can grow with L	mean recursion saturates in L (Thm. 5)
Centered scale	correlation propagation	gap $\Theta^{(L)}(1) - \Theta^{(L)}(\rho) \asymp 1/(rL)$ (Thm. 5)

Table 1: Depth/rank effects in the kernel regime. MLP: fully trained behavior; RF-LR: explicit $1/r$ bottleneck, saturation and early-layer suppression.

A.5. Comparison table: depth/rank effects in the kernel regime

A.6. Techniques and proof sketch

We highlight the main proof ingredients and how they combine.

Recursion and closed form. The RF-LR NTK recursion (Theorem 3) follows from a layer-wise decomposition of the gradient inner product into bias, readout, and backpropagated contributions, together with conditional law of large numbers arguments under the sequential infinite-width limit [Jacot et al. \(2018\)](#). Iterating the recursion yields the explicit $2L$ -term expansion (Corollary 4).

Microscopic decoupling and concentration. For three layers, the empirical kernel depends on the sample cosine and norm product of rank- r Gaussian projections. Rotational invariance yields Fisher–Kibble decoupling (Lemma 9), separating angular and radial fluctuations [Fisher \(1915\)](#); [Hotelling \(1953\)](#); [Kibble \(1941\)](#). Concentration of Gaussian norms and sample correlations then yields sub-Gaussian deviations in r for the empirical kernel (Theorem 27 and Corollary 28).

RKHS identification via endpoint expansions. The RKHS of zonal kernels on the sphere is controlled by the Puiseux behavior near $\rho = \pm 1$ [Bietti and Bach \(2021\)](#). We compute the mean three-layer kernel’s endpoint expansion by combining Fisher’s density with a hypergeometric connection formula, showing that the leading $t^{1/2}$ exponent is preserved after taking expectations, hence the mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 14).

Depth scaling and effective depth. Depth dependence is controlled by correlation propagation and by products of derivative kernels. In the bottleneck regime, the RF-LR recursion carries explicit $1/r$ prefactors at each bottleneck layer, and for ReLU one has $\sup_\rho \tilde{\Sigma}^{(\ell)}(\rho) \leq 1/2$, which yields exponential suppression of early-layer contributions as depth increases (Section 4). A refined mean-kernel statement is obtained by analyzing the deterministic proxy recursion (Theorem 5).

Future work: quadratic random-matrix scaling. Establishing bulk spectral laws for the *empirical* RF-LR NTK in a quadratic scaling regime $n \rightarrow \infty$ requires additional resolvent/local-law inputs. While such bulk laws are now available for shallow two-layer NTKs via resolvent replacement and Gaussian-equivalence machinery [Benigni and Paquette \(2025\)](#), adapting these methods to the present low-rank random-feature structure and to depth $L \geq 2$ is left for future work.

Appendix B. Proofs of NTK Recursions

B.1. Proof of Theorem 3: Infinite-width NTK recursion

Proof sketch. Decompose the NTK at layer ℓ into (i) a bias contribution (constant +1), (ii) Term 2: gradient inner products for the layer- ℓ readouts $A^{(\ell)}$, and (iii) Term 1: backpropagated NTK from previous layers times the Jacobian metric. Apply the Law of Large Numbers conditionally on $\mathbf{h}^{(\ell-1)}$ as $n_\ell \rightarrow \infty$; the sequential limit ensures lower layers have already converged.

Proof We prove the recursion by decomposing the NTK at layer ℓ into contributions of the trainable parameters up to layer ℓ , and invoking the Law of Large Numbers (LLN) under the sequential infinite-width limit. Let $f^{(\ell)}$ denote the network output truncated at layer ℓ . For finite width, the (random) NTK reads

$$\Theta_{\text{rand}}^{(\ell)}(x, x') = \underbrace{\left(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x) \right)^\top \left(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x') \right)}_{\text{Term 1}} + \underbrace{\sum_{j=1}^{n_\ell} \frac{\partial f^{(\ell)}(x)}{\partial A_j^{(\ell)}} \frac{\partial f^{(\ell)}(x')}{\partial A_j^{(\ell)}}}_{\text{Term 2}}. \quad (15)$$

Bias parameters. We include a trainable layer bias $c^{(\ell)}$ (initialized at zero). The gradient feature with respect to $c^{(\ell)}$ is constant, so it contributes an additive constant term to the NTK; under our normalization this contribution is 1. This yields the additive +1 term in the recursion.

Term 2 (layer- ℓ output weights $A^{(\ell)}$). By the forward pass, $\partial f^{(\ell)}(x) / \partial A_j^{(\ell)} = \frac{\sigma_A}{\sqrt{n_\ell}} \boldsymbol{\sigma}(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x))$. Therefore

$$\text{Term 2} = \frac{\sigma_A^2}{n_\ell} \sum_{j=1}^{n_\ell} \boldsymbol{\sigma}(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)) \boldsymbol{\sigma}(\mathbf{w}_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x')). \quad (16)$$

Conditionally on $\mathbf{h}^{(\ell-1)}$, the summands are i.i.d. over $\mathbf{w}_j^{(\ell)}$. As $n_\ell \rightarrow \infty$ (with lower layers already taken to their limits), LLN yields

$$\text{Term 2} \xrightarrow{p} \sigma_A^2 \Sigma^{(\ell)}(x, x'). \quad (17)$$

Term 1 (previous layers). By the chain rule,

$$\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x) = \left(\frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)} \right) \nabla_{\theta^{(\ell-1)}} \mathbf{h}^{(\ell-1)}(x). \quad (18)$$

Hence Term 1 factorizes as $\Theta_{\text{rand}}^{(\ell-1)}(x, x')$ multiplied by a Jacobian metric:

$$\text{Term 1} = \Theta_{\text{rand}}^{(\ell-1)}(x, x') \cdot \left\langle \frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)}, \frac{\partial f^{(\ell)}(x')}{\partial \mathbf{h}^{(\ell-1)}(x')} \right\rangle. \quad (19)$$

Taking the inner product and applying LLN as $n_\ell \rightarrow \infty$ yields

$$\left\langle \frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)}, \frac{\partial f^{(\ell)}(x')}{\partial \mathbf{h}^{(\ell-1)}(x')} \right\rangle \xrightarrow{p} \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (20)$$

Therefore,

$$\text{Term 1} \xrightarrow{p} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (21)$$

Combining the bias contribution with the two limits completes the recursion. For bottleneck layers $\ell \geq 2$, the Jacobian and output contributions scale with $1/r$ (from the r -dimensional bottleneck); we absorb σ_A^2 into the normalization. Thus

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \quad \Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \sigma_A^2 \Sigma^{(\ell)}(x, x') \quad (\ell \geq 2) \quad (22)$$

The sequential limit assumption ensures that lower-layer random objects have already converged when taking the next-layer limit, justifying the conditional LLN applications. $\blacksquare$

B.2. Proof of Theorem 2: Gaussian process composition

Proof sketch. By induction on ℓ : for $\ell = 0$ the output is deterministic; for $\ell \geq 1$, conditionally on $\mathbf{h}^{(\ell-1)}$ the forward pass is a sum of i.i.d. terms over j , so the multivariate CLT yields Gaussian convergence. The covariance matches the base kernel $\Sigma^{(\ell)}$ by definition.

Proof We prove by induction on ℓ that $\mathbf{h}^{(\ell)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell)}$.

Base case $\ell = 0$. For $\ell = 0$, we have $\mathbf{h}^{(0)}(x) = x$, which is deterministic and trivially a Gaussian process.

Inductive step. Assume that $\mathbf{h}^{(\ell-1)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell-1)}$. Consider the forward pass at layer ℓ :

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma} \left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x) \right) + c^{(\ell)}. \quad (23)$$

For any finite collection of inputs $\{x_1, \dots, x_m\}$, conditionally on $\mathbf{h}^{(\ell-1)}$, the terms $A_j^{(\ell)} \boldsymbol{\sigma} \left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x_i) \right)$ are independent across j and identically distributed. Since $A_j^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2)$ and $w_j^{(\ell)}$ are frozen Gaussian draws, the conditional distribution of each term is centered with finite variance (and $c^{(\ell)}$ is deterministic at initialization).

By the multivariate Central Limit Theorem, as $n_\ell \rightarrow \infty$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian. The covariance is given by

$$\mathbb{E} \left[\mathbf{h}^{(\ell)}(x_i) \mathbf{h}^{(\ell)}(x_j)^\top \right] = \mathbb{E}_w \left[\boldsymbol{\sigma} \left(w^\top \mathbf{h}^{(\ell-1)}(x_i) \right) \boldsymbol{\sigma} \left(w^\top \mathbf{h}^{(\ell-1)}(x_j) \right)^\top \right] = \Sigma^{(\ell)}(x_i, x_j), \quad (24)$$

where the expectation is taken over the frozen weights w and the limiting Gaussian process $\mathbf{h}^{(\ell-1)}$. The last equality follows from the definition of the base kernel in Definition 1.

The sequential infinite-width limit ensures that $\mathbf{h}^{(\ell-1)}$ has already converged to its Gaussian process limit when taking the limit for layer ℓ , justifying the application of the CLT conditionally on $\mathbf{h}^{(\ell-1)}$. $\blacksquare$

B.3. Proof of three-layer NTK Corollary

Proof sketch. Apply the recursion (Theorem 3) for $\ell = 1$ and $\ell = 2$; for $\ell = 1$ use $\Theta^{(0)} = 0$; for $\ell = 2$ substitute $\Theta^{(1)} = 1 + \Sigma^{(1)}$. Under homogeneity, the mean form involves expectations over Fisher (angular) and Kibble (radial); as $r \rightarrow \infty$ these concentrate, recovering K_∞ .

Proof Apply Theorem 3 for $\ell = 1$ and $\ell = 2$. For $\ell = 1$, since $\Theta^{(0)} = 0$, the recursion gives:

$$\Theta^{(1)}(x, x') = 1 + \Theta^{(0)}(x, x') \cdot \dot{\Sigma}^{(1)}(x, x') + \Sigma^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'). \quad (25)$$

For $\ell = 2$, substitute $\Theta^{(1)} = 1 + \Sigma^{(1)}$ into the recursion:

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x'). \quad (26)$$

This is the stated three-layer recursion with the bias contribution.

Mean explicit form (low-rank). Under the homogeneous ReLU setting on the sphere, the mean three-layer NTK (with $1/r$ factors) is

$$\tilde{\Theta}^{(2)}(\rho) = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \mathbb{E} \left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi} \right] + \frac{1}{r} \mathbb{E} \left[\Sigma^{(1)}(\hat{\rho}_r) \|x_1\| \|y_1\| \right], \quad (27)$$

where the expectations are over Fisher and Kibble. As $r \rightarrow \infty$, $\hat{\rho}_r \rightarrow \rho$ and $\|x_1\| \|y_1\| / r \rightarrow 1$, so the limit $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ is recovered (without the $1/r$ prefactors, since the low-rank scaling is absorbed in the limit). ■

B.4. Proof of General L -layer NTK Corollary

Remark 17 (Explicit $2L$ -term unrolling) *Equivalently, expanding the recursion yields an explicit linear number of terms (namely $2L$). For example, for $L = 3$:*

$$\begin{aligned} \Theta^{(3)}(x, x') &= 1 + \frac{1}{r} \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \dot{\Sigma}^{(3)} \\ &\quad + \frac{1}{r^2} \Sigma^{(1)} \cdot \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(3)}. \end{aligned} \quad (28)$$

Each bottleneck layer contributes a factor $1/r$; layer 1 has no such factor. In total, there are $2L$ terms for depth L .

Proof sketch. By induction on L : for $L = 1$ the recursion gives $\Theta^{(1)} = 1 + \Sigma^{(1)}$. For $L \geq 2$, expand $\Theta^{(L)}$ via the recursion into a bias term and two paths (derivative-propagated and fresh-base); each path contributes L terms with the appropriate $1/r$ factors, yielding $2L$ terms in total.

Proof We prove the explicit form by induction on L , accounting for the $+1$ bias terms in the recursion.

Base case $L = 1$. From Theorem 3, $\Theta^{(1)} = 1 + \Sigma^{(1)}$. The formula with $L = 1$ gives:

$$\Theta^{(1)}(x, x') = \sum_{\ell=1}^1 \left(\prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^1 \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x'). \quad (29)$$

For the first sum: when $\ell = 1$, $\prod_{k=2}^1 \dot{\Sigma}^{(k)} = 1$ (empty product). For the second sum: when $\ell = 1$, $\Sigma^{(1)} \cdot \prod_{k=2}^1 \dot{\Sigma}^{(k)} = \Sigma^{(1)} \cdot 1 = \Sigma^{(1)}$. Therefore, $\Theta^{(1)} = 1 + \Sigma^{(1)}$, which matches the recursion.

Inductive step. Assume the formula holds for depth $L - 1$:

$$\Theta^{(L-1)}(x, x') = \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^{L-1} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x'). \quad (30)$$

By Theorem 3, for $L \geq 2$,

$$\Theta^{(L)}(x, x') = 1 + \frac{1}{r} \Theta^{(L-1)}(x, x') \cdot \dot{\Sigma}^{(L)}(x, x') + \frac{1}{r} \Sigma^{(L)}(x, x'). \quad (31)$$

Substituting the inductive hypothesis and distributing the $1/r$ factor yields the stated closed form; the first sum gains a new term 1 (from $\ell = L$) and each existing term is multiplied by $(1/r)\dot{\Sigma}^{(L)}$, while the second sum gains $(1/r)\Sigma^{(L)}$ and each existing term is multiplied by $(1/r)\dot{\Sigma}^{(L)}$. The first term 1 corresponds to $\ell = L$ in the first sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). The last term $\Sigma^{(L)}$ corresponds to $\ell = L$ in the second sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). Therefore:

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (32)$$

completing the induction. The total number of terms is $L + L = 2L$, which grows linearly with depth. $\blacksquare$

B.5. Reference: full-width MLP limiting NTK at the edge of chaos

Proposition 18 (Closed form for full-width MLP EOC NTK) *Let $l \geq 1$ and consider an infinitely wide l -layer MLP with (a, b) -ReLU activation initialized at the edge of chaos as in [Terjék and González-Sánchez \(2025\)](#). Denote by ϱ the corresponding cosine map and by $\rho_1(x, x')$ the input cosine similarity. Then the limiting NTK satisfies*

$$K_{\infty}(x, x') = \|x\| \|x'\| \left(\sum_{k=1}^l \varrho^{\circ(k-1)}(\rho_1(x, x')) \prod_{k'=k}^{l-1} \varrho'(\varrho^{\circ(k'-1)}(\rho_1(x, x'))) \right) I_{m_l},$$

with the convention that the empty product (for $k = l$) equals 1.

Proof sketch. The fully trained MLP NTK at EOC admits an explicit closed form in terms of the cosine map ϱ and its derivative; we refer to [Terjék and González-Sánchez \(2025\)](#) for the derivation.

Proof This is [Terjék and González-Sánchez \(2025\)](#); we include it for completeness. $\blacksquare$

Appendix C. Proofs of low-rank NTK RKHS

C.1. Background and statements for Section 5

RKHS viewpoint in the NTK regime. In the NTK regime, training dynamics linearize:

$$f_t(x) \approx f_0(x) + \langle \nabla_{\theta} f_0(x), \theta_t - \theta_0 \rangle.$$

The learned function stays in the RKHS induced by the (limiting) kernel Θ , with norm

$$\|f\|_{\mathcal{H}_\Theta}^2 = \langle f, \Theta^{-1}f \rangle_{L^2}.$$

For RF-LR with frozen random features, universal approximation holds due to full support of the frozen Gaussian draws $w^{(\ell)}$, so any function in the Barron space [E et al. \(2021\)](#) can be approximated with error $O(1/\sqrt{N})$. Classical NTK analysis suggests that feature weights move little during training; RF-LR enforces this by design by freezing the feature weights, which isolates kernel-regime behavior without an additional weight-deviation analysis.

Mean RKHS and random features. Each realization ω of the random features yields a kernel $K_\omega(x, x') = \Theta_\omega^{(2)}(x, x')$ and an induced RKHS $\mathcal{H}_\omega$. The mean kernel $\tilde{K}(x, x') = \mathbb{E}_\omega[K_\omega(x, x')]$ is deterministic; its RKHS $\mathcal{H}_{\tilde{K}}$ is the RKHS of $\tilde{K}$. It is not $\mathbb{E}[\mathcal{H}_\omega]$ (expectation of Hilbert spaces is undefined). For each ω , the feature map $\varphi_\omega(x) = K_\omega(\cdot, x) \in \mathcal{H}_\omega$ is random, whereas the canonical feature map of the mean kernel $\varphi_{\tilde{K}}(x) = \tilde{K}(\cdot, x) \in \mathcal{H}_{\tilde{K}}$ is deterministic. The mean RKHS describes the typical function space obtained after averaging over random feature realizations.

Remark (application of Bietti–Bach). Once the mean kernel’s Puiseux endpoint behavior is established (Appendix C.9), the RKHS equivalence in Corollary 14 follows directly from the characterization theorem of [Bietti and Bach \(2021\)](#).

Corollary 19 (Mean NTK over Fisher–Kibble has same RKHS: three-layer case) *Under Assumption 1 and the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the mean three-layer NTK $\tilde{\Theta}^{(2)}$ (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow ReLU kernel. In particular, the RKHSs coincide as sets with equivalent norms.*

Proof: See Appendix C.9.

Main RKHS result (stated in the main text). Corollary 14 is stated in Section 5. It follows by combining the Puiseux endpoint analysis in Appendix C.9 with the RKHS characterization of [Bietti and Bach \(2021\)](#).

C.2. Discussion on Fisher and Kibble Distributions

Lemma 20 (Fisher–Kibble decoupling) *Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#) [Hotelling \(1953\)](#), centered at the true correlation ρ .
- (u, v) follow Kibble’s [Kibble \(1941\)](#) bivariate Gamma (chi-square) law.
- Angular and radial parts are independent: $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Remark 21 (Fisher–Kibble decoupling and independence) *By rotational invariance, the empirical correlation and squared norms satisfy*

$$\rho_1 \sim \text{Fisher}(\rho, r), \quad (u, v) \sim \text{Kibble}(r, \rho), \quad \text{and} \quad p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v).$$

Homogeneity and factorization (why angular and radial parts decouple). For positively 1-homogeneous activations (Assumption 1), the first-layer base kernel separates radial and angular dependence: for all $x, x' \neq 0$,

$$\Sigma^{(1)}(x, x') = \|x\| \|x'\| \tilde{\Sigma}^{(1)}(\rho), \quad \rho = \frac{\langle x, x' \rangle}{\|x\| \|x'\|},$$

where $\tilde{\Sigma}^{(1)}$ is a scalar function on $[-1, 1]$ (for ReLU it is the standard arc-cosine kernel). Under isotropic Gaussian projections, rotational invariance implies that the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ depends only on the angular part of the projected pair, while the norms $(\|x_1\|, \|y_1\|)$ depend only on the radial part; consequently ρ_1 is independent of $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ and the joint law factorizes as stated above. The Fisher distribution density [Fisher \(1915\)](#) is

$$p(\rho_1 \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-\rho_1^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho\rho_1)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho\rho_1}{2}\right), \quad r > 2.$$

For large r , the inverse Gudermann function applied to the complementary angle is approximately normal:

$$\text{gd}^{-1}\left(\frac{\pi}{2} - \arccos(\rho_1)\right) \approx \mathcal{N}\left(\text{arctanh}(\rho), \frac{1}{r-3}\right), \quad \text{hence} \quad \text{Var}(\rho_1) = O(1/r).$$

Kibble's [Kibble \(1941\)](#) bivariate chi-square law for $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ has density

$$f(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} \exp\left(-\frac{u+v}{2(1-\rho^2)}\right)}{\Gamma(r/2) (2(1-\rho^2))^{\frac{r}{2}+1} \rho^{\frac{r/2-1}{2}}} I_{\frac{r}{2}-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right), \quad u, v \geq 0,$$

where I_ν is the modified Bessel function of the first kind. The moments are

$$\mathbb{E}[u] = \mathbb{E}[v] = r, \quad \text{Var}(u) = \text{Var}(v) = 2r, \quad \text{Cov}(u, v) = 2r \rho^2.$$

Defining $w_r = \sqrt{uv}/r$, concentration results imply $\mathbb{E}[w_r] \rightarrow 1$ and $\text{Var}(w_r) = O(1/r)$.

Compact three-layer empirical NTK representation. In the three-layer low-rank setting, denote

$$\rho_1 = \cos \angle(\mathbf{h}^{(1)}(x), \mathbf{h}^{(1)}(x')), \quad w_r = \frac{\|\mathbf{h}^{(1)}(x)\| \|\mathbf{h}^{(1)}(x')\|}{r}.$$

Then the empirical three-layer NTK admits the compact form

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(\rho_1) \left(1 - \frac{\arccos(\rho_1)}{\pi}\right) + \frac{1}{r} w_r \Sigma^{(1)}(\rho_1),$$

with $\Sigma^{(1)}(u) = \frac{1}{\pi} (\sqrt{1-u^2} + u(1 - \arccos u))$.

Proof sketch. The rank- r projections $x_1 = Px$, $y_1 = Py$ form a ρ -correlated Gaussian pair. By rotational invariance, the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ (angular part) and the squared norms $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ (radial part) are independent. The angular part follows Fisher's distribution; the radial part follows Kibble's bivariate chi-square law.

Proof Let $x, y \in \mathbb{R}^d$ be fixed input vectors with correlation $\rho = \langle x, y \rangle / (\|x\| \|y\|)$. Consider a rank- r random projection matrix $P \in \mathbb{R}^{r \times d}$ with entries $P_{ij} \sim \mathcal{N}(0, 1/d)$ i.i.d., so that $x_1 = Px$ and $y_1 = Py$ are the projected vectors.

Since P has i.i.d. Gaussian entries, the projected vectors $x_1, y_1 \in \mathbb{R}^r$ form a ρ -correlated bivariate Gaussian pair. Specifically, for each component $i = 1, \dots, r$, the pairs (x_{1i}, y_{1i}) are i.i.d. with joint distribution

$$\begin{pmatrix} x_{1i} \\ y_{1i} \end{pmatrix} \sim \mathcal{N} \left(\mathbf{0}, \begin{pmatrix} \|x\|^2/d & \rho \|x\| \|y\|/d \\ \rho \|x\| \|y\|/d & \|y\|^2/d \end{pmatrix} \right). \quad (33)$$

By rotational invariance of the Gaussian projection and the fact that the correlation ρ is preserved under orthogonal transformations, we can assume without loss of generality that the covariance structure factors into angular and radial components.

Angular part (Fisher distribution). The empirical correlation is defined as

$$\rho_1 = \frac{\langle x_1, y_1 \rangle}{\|x_1\| \|y_1\|} = \frac{\sum_{i=1}^r x_{1i} y_{1i}}{\sqrt{\sum_{i=1}^r x_{1i}^2} \sqrt{\sum_{i=1}^r y_{1i}^2}}. \quad (34)$$

This is the sample correlation coefficient of r pairs of correlated Gaussian random variables. Fisher [Fisher \(1915\)](#); [Hotelling \(1953\)](#) showed that the distribution of ρ_1 depends only on the true correlation ρ and the sample size r , with density given in Remark 20. The key observation is that ρ_1 is a function purely of the angles between the projected vectors, independent of their magnitudes.

Radial part (Kibble distribution). The squared norms $u = \|x_1\|^2 = \sum_{i=1}^r x_{1i}^2$ and $v = \|y_1\|^2 = \sum_{i=1}^r y_{1i}^2$ are sums of correlated chi-square variables. Since (x_{1i}, y_{1i}) are ρ -correlated Gaussians, the bivariate distribution of (u, v) follows Kibble's [Kibble \(1941\)](#) bivariate chi-square law. The joint density involves the modified Bessel function $I_\nu(z)$, which appears because the dot product $\langle x_1, y_1 \rangle = \sum_{i=1}^r x_{1i} y_{1i}$ can be expressed as a weighted sum of products of correlated normals, whose distribution relates to Bessel functions through the generating function of the bivariate chi-square distribution.

Specifically, the joint characteristic function of (u, v) is

$$\phi(s, t) = \mathbb{E}[e^{i(su+tv)}] = (1 - 2i(1 - \rho^2)(s + t) - 4\rho^2 st)^{-r/2}, \quad (35)$$

and the inverse Fourier transform yields Kibble's density, which contains the modified Bessel function $I_{\frac{r}{2}-1}$ as stated in Remark 20. The Bessel function arises from the integral representation of the bivariate chi-square density when the correlation $\rho \neq 0$.

Independence of angular and radial parts. Rotational invariance of isotropic Gaussian projections implies that $\rho_1 = \cos \angle(x_1, y_1)$ and the norms $(\|x_1\|, \|y_1\|)$ are independent, yielding the factorization $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) \cdot p_{\text{Kibble}}(u, v)$. This is the classical independence of the sample correlation from sample variances in bivariate normal data. $\blacksquare$

C.3. Proof of kernel CLT (asymptotic normality of low-rank kernel)

By Fisher–Kibble decoupling (Lemma 9), the angular and radial parts are independent; a joint CLT and the Delta method at $(\rho, 1)$ yield the stated asymptotic normality. The variance $\sigma_{\text{tot}}^2(\rho)$ is as in the main text; details parallel the concentration argument in Appendix C.7.

C.4. Proof of spike–bulk spectral limit

Under the random-matrix scaling and data assumptions stated in the spectra section, the empirical spectrum of the Gram matrix built from K_∞ converges to a deformed Marchenko–Pastur law with a rank-one spike and bulk support as stated; the result follows from the inner-product kernel framework of El Karoui (2010).

C.5. Open directions: deep mean NTK recursion and effective depth

The deterministic proxy (Definition D.4) analyzes the scalar recursion along the mean path $\rho_k = \varrho^{\circ(k-1)}(\rho_1)$. The actual network uses the *random* correlation chain ρ_ℓ with $O(1/\sqrt{r})$ per-layer fluctuations (Appendix A.4). A rigorous link requires: (i) propagating the Fisher-type concentration $\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1))^2] \leq C/r$ through depth to show $\Theta_{\text{random}}^{(L)} - \Theta_{\text{proxy}}^{(L)} = O_P(\sqrt{L/r})$ or similar; (ii) a rigorous recursion for $\tilde{\Theta}^{(\ell)}$ with explicit covariance bounds; (iii) extension of Fisher–Kibble decoupling to nested layers to control mixed terms such as $\mathbb{E}[\Theta^{(\ell-1)} \dot{\Sigma}^{(\ell)}]$ (where dependence through the deep GP chain prevents naive factorization). The exponential suppression of early-layer contributions suggests that very deep MMNNs may exhibit an “effective depth” phenomenon similar to ResNets, where only the top layers meaningfully contribute to the NTK.

Lemma 22 (Nonasymptotic concentration of Gaussian sample cosine) *Let $r \geq 1$ and let $(X, Y) \in \mathbb{R}^r \times \mathbb{R}^r$ have i.i.d. coordinates $\{(X_i, Y_i)\}_{i=1}^r$, each distributed as a centered bivariate normal with $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2] = 1$ and $\mathbb{E}[X_i Y_i] = \rho \in (-1, 1)$. Define the empirical cosine (sample correlation without centering)*

$$\hat{\rho}_r = \frac{\langle X, Y \rangle}{\|X\| \|Y\|}.$$

Then there exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq 6 \exp(-c r t^2), \quad \mathbb{E}[(\hat{\rho}_r - \rho)^2] \leq \frac{C}{r}.$$

Proof sketch. Write $\hat{\rho}_r = S_{xy} / \sqrt{S_{xx} S_{yy}}$ where S_{xx}, S_{yy}, S_{xy} are sample averages. Each of $S_{xx} - 1, S_{yy} - 1, S_{xy} - \rho$ is sub-exponential; Bernstein’s inequality yields tail bounds. On the event that S_{xx}, S_{yy} are close to 1, bound $|\hat{\rho}_r - \rho|$ via a ratio expansion; union bound over the three tails gives the stated rate.

Proof Write

$$S_{xx} = \frac{1}{r} \|X\|^2, \quad S_{yy} = \frac{1}{r} \|Y\|^2, \quad S_{xy} = \frac{1}{r} \langle X, Y \rangle, \quad \text{so that} \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}.$$

Each of $S_{xx} - 1, S_{yy} - 1$, and $S_{xy} - \rho$ is an average of i.i.d. centered sub-exponential random variables (since $X_i^2 - 1, Y_i^2 - 1$, and $X_i Y_i - \rho$ have finite ψ_1 norms for Gaussian data). Therefore,

by Bernstein's inequality, there exist absolute constants $c_0 > 0$ and $C_0 < \infty$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{yy} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{xy} - \rho| \geq t) \leq 2e^{-c_0 r t^2}.$$

On the event $E_t = \{|S_{xx} - 1| \leq t, |S_{yy} - 1| \leq t\}$ with $t \leq 1/2$, we have $S_{xx}, S_{yy} \in [1/2, 3/2]$, hence

$$\left| \frac{1}{\sqrt{S_{xx}S_{yy}}} - 1 \right| \leq C_0(|S_{xx} - 1| + |S_{yy} - 1|) \leq 2C_0 t$$

for an absolute constant C_0 . Using

$$\hat{\rho}_r - \rho = \frac{S_{xy} - \rho}{\sqrt{S_{xx}S_{yy}}} + \rho \left(\frac{1}{\sqrt{S_{xx}S_{yy}}} - 1 \right),$$

we obtain on E_t the bound $|\hat{\rho}_r - \rho| \leq C_1(|S_{xy} - \rho| + t)$ for an absolute constant C_1 . Taking $t \asymp t'$ and applying a union bound over the three Bernstein tails yields $\mathbb{P}(|\hat{\rho}_r - \rho| \geq t') \leq 6e^{-c r t'^2}$ for $t' \in (0, 1)$ and absolute $c > 0$. The second-moment bound follows by integrating the tail bound: $\mathbb{E}[(\hat{\rho}_r - \rho)^2] = \int_0^\infty 2u \mathbb{P}(|\hat{\rho}_r - \rho| \geq u) du \leq C/r$. $\blacksquare$

C.6. Concentration of the random Gram matrix around the proxy

Let $\hat{K}$ be the $n \times n$ random Gram matrix with entries $\hat{K}_{ij} = \hat{\Theta}_r^{(L)}(x_i, x_j)$ (the empirical RF-LR NTK over the random correlation chain), and let K_{proxy} be the deterministic proxy Gram matrix with entries $(K_{\text{proxy}})_{ij} = \Theta^{(L)}(\rho_{1,ij})$ (Definition D.4). Under the same per-layer sub-Gaussian concentration as in Appendix A.4, $\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1))^2] \leq C/r$ and $\mathbb{P}(|\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1)| \geq t) \leq 6 \exp(-c r t^2)$, the kernel recursion is Lipschitz in the correlation path. Propagating the per-step bounds through L layers (e.g. by a union bound over the L steps and the fact that $\Theta^{(k)}$ is Lipschitz in ρ with constant $O(1)$), one obtains that for each pair (i, j) , $|\hat{K}_{ij} - (K_{\text{proxy}})_{ij}| = O_P(\sqrt{L/r})$. A union over the n^2 pairs and the inequality $\|A\|_{\text{op}} \leq \|A\|_{\text{F}} \leq n \max_{i,j} |A_{ij}|$ yield the following. The bound in Proposition 22 is *sketch-based*; a full proof with explicit constants and rate is given only for the equicorrelated case (Theorem 25).

Proposition 23 (Proxy–empirical Gram matrix concentration) *Fix $n, L \geq 1, r > 1$, and inputs $x_1, \dots, x_n$ with pairwise cosine similarities $\rho_{1,ij} \in (-1, 1)$. Let $\hat{K}$ and K_{proxy} be as above. There exist constants $C, c > 0$ (depending on L, n and the $\rho_{1,ij}$) such that for any $\epsilon \in (0, 1)$,*

$$\mathbb{P}\left(\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} \geq \epsilon\right) \leq C \exp\left(-c \frac{r \epsilon^2}{L n^2}\right).$$

In particular, for $r \rightarrow \infty$ with n, L fixed, $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = o_P(1)$. If in addition the proxy Gram matrix has minimum eigenvalue $\lambda_{\min}(K_{\text{proxy}}|_{\mathbf{1}^\perp}) \geq \gamma > 0$, then for r large enough so that $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} < \gamma/2$ with high probability, Weyl's inequality gives $|\lambda_{\min}(\hat{K}|_{\mathbf{1}^\perp}) - \lambda_{\min}(K_{\text{proxy}}|_{\mathbf{1}^\perp})| \leq \gamma/2$, so the condition number of $\hat{K}$ restricted to $\mathbf{1}^\perp$ is within a constant factor of that of K_{proxy} .

Proof sketch. Per-layer concentration (Lemma 21) and Lipschitz dependence of the scalar recursion on ρ imply that each entry $\hat{K}_{ij}$ differs from $(K_{\text{proxy}})_{ij}$ by $O_P(\sqrt{L/r})$. Union bound over n^2 pairs with sub-Gaussian tails gives $\max_{i,j} |\hat{K}_{ij} - (K_{\text{proxy}})_{ij}| = O_P(\sqrt{L \log(n)/r})$. Hence $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} \leq n \|\hat{K} - K_{\text{proxy}}\|_{\text{max}} = O_P(n\sqrt{L \log(n)/r})$, and the stated exponential tail follows. The condition-number comparison is by Weyl's inequality for the eigenvalues of symmetric matrices. For general (non-euicorrelated) datasets this remains a proof sketch; a rigorous operator-norm bound with explicit rate is given only for the euicorrelated case in Theorem 25.

Euicorrelated case: reduction to two scalars and rigorous bound on $\mathbf{1}^\perp$. In the euicorrelated case ($\rho_{1,ij} = \rho_0$ for all $i \neq j$, $\rho_{1,ii} = 1$), the proxy Gram matrix has the form $K_{\text{proxy}} = \Theta^{(L)}(1)I_n + \Theta^{(L)}(\rho_0)(\mathbf{1}\mathbf{1}^\top - I_n)$. By symmetry of the data and the network, the random $\hat{K}$ has the same structure: $\hat{K}_{ii} = \hat{K}_{11}$ and $\hat{K}_{ij} = \hat{K}_{12}$ for $i \neq j$. Hence the deviation matrix $E\hat{K} - K_{\text{proxy}}$ is

$$E = (\hat{K}_{11} - \Theta^{(L)}(1))I_n + (\hat{K}_{12} - \Theta^{(L)}(\rho_0))(\mathbf{1}\mathbf{1}^\top - I_n),$$

with eigenvalues $\lambda_1 = (\hat{K}_{11} - \Theta^{(L)}(1)) + (n-1)(\hat{K}_{12} - \Theta^{(L)}(\rho_0))$ on $\mathbf{1}$ and $\lambda_\perp = (\hat{K}_{11} - \Theta^{(L)}(1)) - (\hat{K}_{12} - \Theta^{(L)}(\rho_0))$ on $\mathbf{1}^\perp$ (multiplicity $n-1$), so $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = \max(|\lambda_1|, |\lambda_\perp|)$. The following theorem gives a rigorous $O_P(L/r)$ bound for the restriction to $\mathbf{1}^\perp$; the full operator norm without n -dependence remains open (see Remark 26).

Lemma 24 (Path-wise kernel and sensitivity) *Let $s(\rho)$ and $\dot{s}(\rho)$ be the scalar ReLU base and derivative kernels as in (83). For $k \geq 1$ and $\rho_1, \dots, \rho_k \in (-1, 1)$, define $\Phi^{(1)}(\rho_1) = 1 + s(\rho_1)$ and for $k \geq 2$,*

$$\Phi^{(k)}(\rho_1, \dots, \rho_k) = 1 + \frac{1}{r} \Phi^{(k-1)}(\rho_1, \dots, \rho_{k-1}) \dot{s}(\rho_k) + \frac{1}{r} s(\rho_k).$$

Then $\Phi^{(k)}$ is the same function as $\Theta^{(k)}$ when evaluated along the deterministic path $\rho_j = \varrho^{\circ(j-1)}(\rho_1)$. There exist constants $C_s, C_\Phi < \infty$ (depending only on the ReLU kernels) such that for all $2 \leq \ell \leq k \leq L$ and all $\rho_1, \dots, \rho_k \in [-1 + \delta, 1 - \delta]$ with $\delta \in (0, 1)$,

$$\left| \frac{\partial \Phi^{(k)}}{\partial \rho_\ell} \right| \leq C_\Phi \left(\frac{1}{r} \right)^{k-\ell}.$$

In particular, $\sum_{\ell=2}^L \left(\frac{\partial \Phi^{(L)}}{\partial \rho_\ell} \right)^2 \leq C_\Phi^2 \frac{1-(1/r)^{2(L-1)}}{1-(1/r)^2} \leq \bar{C}_L$ and $\sum_{\ell=2}^L \left| \frac{\partial \Phi^{(L)}}{\partial \rho_\ell} \right| \leq C_\Phi \sum_{t=0}^{L-2} (1/r)^t \leq \bar{C}_L''$ for some $\bar{C}_L, \bar{C}_L'' = O(1)$ when $r \geq 2$.

Proof The recursion for $\Phi^{(k)}$ coincides with (83), so $\Theta^{(k)}(\rho_1) = \Phi^{(k)}(\rho_1, \varrho(\rho_1), \dots, \varrho^{\circ(k-1)}(\rho_1))$. Differentiating, $\frac{\partial \Phi^{(k)}}{\partial \rho_k} = \frac{1}{r} \Phi^{(k-1)} \dot{s}'(\rho_k) + \frac{1}{r} s'(\rho_k)$. On $[-1 + \delta, 1 - \delta]$, $|\dot{s}'|$ and $|s'|$ are bounded by some C_s . Since $\Phi^{(k-1)}$ is bounded (the recursion yields $\Phi^{(k)} \in [1, 1+O(1)]$ on compact ρ), we have $\left| \frac{\partial \Phi^{(k)}}{\partial \rho_k} \right| \leq C'_s(1/r)$ for a constant C'_s . For $\ell < k$, $\frac{\partial \Phi^{(k)}}{\partial \rho_\ell} = \frac{1}{r} \frac{\partial \Phi^{(k-1)}}{\partial \rho_\ell} \dot{s}(\rho_k)$. By induction, $\left| \frac{\partial \Phi^{(k)}}{\partial \rho_\ell} \right| \leq C'_s \prod_{j=\ell+1}^k \frac{1}{r} |\dot{s}(\rho_j)| \leq C_\Phi (1/r)^{k-\ell}$, with $C_\Phi = C'_s (\sup |\dot{s}|)^{L-1} \leq C'_s/2^{L-1}$. The sum bound follows from $\sum_{\ell=2}^L (1/r)^{2(L-\ell)} = \sum_{t=0}^{L-2} (1/r)^{2t} \leq \frac{1}{1-(1/r)^2} \leq 2$ for $r \geq 2$, so $\sum_{\ell=2}^L \left(\frac{\partial \Phi^{(L)}}{\partial \rho_\ell} \right)^2 \leq C_\Phi^2 \cdot 2 = \bar{C}_L$. $\blacksquare$

Lemma 25 (Second-order remainder) Under the same setting as Lemma 23, let $\rho_1 = \rho_0$ be fixed and $(\rho_2, \dots, \rho_L)$ be the random correlation chain with $\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_0))^2] \leq C_0/r$ and $\mathbb{P}(|\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_0)| \geq t) \leq 6 \exp(-c_0 r t^2)$ for all ℓ (Lemma 21). Let $\bar{\rho}_\ell = \varrho^{\circ(\ell-1)}(\rho_0)$. Then the second-order Taylor remainder

$$R_2 = \hat{\Theta}_r^{(L)} - \Theta^{(L)}(\rho_0) - \sum_{\ell=2}^L \frac{\partial \Phi^{(L)}}{\partial \rho_\ell}(\bar{\rho}_2, \dots, \bar{\rho}_L) (\rho_\ell - \bar{\rho}_\ell)$$

satisfies $\mathbb{E}[|R_2|] \leq B_L L/r$ and, for any $u > 0$, $\mathbb{P}(|R_2| \geq u) \leq 2 \exp(-\min(\frac{c_0 r u}{2 B_L L}, \frac{c_0 r u^2}{2 (B_L L)^2}))$ for a constant B_L depending on L and the Hessian of $\Phi^{(L)}$ on $[-1 + \delta, 1 - \delta]^L$.

Proof By Taylor's theorem, $R_2 = \frac{1}{2} \sum_{\ell, m} \frac{\partial^2 \Phi^{(L)}}{\partial \rho_\ell \partial \rho_m}(\xi) (\rho_\ell - \bar{\rho}_\ell)(\rho_m - \bar{\rho}_m)$ for some ξ on the segment between $(\rho_2, \dots, \rho_L)$ and $(\bar{\rho}_2, \dots, \bar{\rho}_L)$. The second derivatives of $\Phi^{(L)}$ are bounded on compact sets (the recursion is polynomial in $s, \bar{s}$ and those are C^∞ on $(-1, 1)$). So $|R_2| \leq \frac{1}{2} H_L \sum_{\ell=2}^L (\rho_\ell - \bar{\rho}_\ell)^2$ for a constant H_L . Hence $\mathbb{E}[|R_2|] \leq \frac{1}{2} H_L \sum_{\ell=2}^L \mathbb{E}[(\rho_\ell - \bar{\rho}_\ell)^2] \leq \frac{1}{2} H_L (L-1) C_0/r \leq B_L L/r$. Each $(\rho_\ell - \bar{\rho}_\ell)^2$ is sub-exponential (from the sub-Gaussian tail of $\rho_\ell - \bar{\rho}_\ell$). By Bernstein's inequality for sums of sub-exponential variables, $\sum_{\ell=2}^L (\rho_\ell - \bar{\rho}_\ell)^2$ has an exponential tail with scale $O(L/r)$; hence $|R_2|$ has the stated tail bound. $\blacksquare$

Proof of Theorem 25. Proof Let $\hat{K}$ and K_{proxy} be as in Proposition 22. Let $E = \hat{K} - K_{\text{proxy}}$. We have $\|E\|_{1^\perp} = |\lambda_\perp| = |\hat{K}_{11} - \Theta^{(L)}(1) - (\hat{K}_{12} - \Theta^{(L)}(\rho_0))|$. For the diagonal, $\rho_1 = 1$ is fixed and the random path $(1, \rho_2^{\text{diag}}, \dots, \rho_L^{\text{diag}})$ has the same per-layer concentration; so $\hat{K}_{11} - \Theta^{(L)}(1)$ admits the same Taylor expansion and bounds as below with ρ_0 replaced by 1. Thus it suffices to prove that both $|\hat{K}_{11} - \Theta^{(L)}(1)|$ and $|\hat{K}_{12} - \Theta^{(L)}(\rho_0)|$ are $O_P(L/r)$; then $|\lambda_\perp| \leq |\hat{K}_{11} - \Theta^{(L)}(1)| + |\hat{K}_{12} - \Theta^{(L)}(\rho_0)| = O_P(L/r)$.

Off-diagonal deviation. Write $\hat{K}_{12} = \Phi^{(L)}(\rho_0, \rho_2, \dots, \rho_L)$ and $\Theta^{(L)}(\rho_0) = \Phi^{(L)}(\rho_0, \bar{\rho}_2, \dots, \bar{\rho}_L)$ with $\bar{\rho}_\ell = \varrho^{\circ(\ell-1)}(\rho_0)$. By Taylor expansion,

$$\hat{K}_{12} - \Theta^{(L)}(\rho_0) = \sum_{\ell=2}^L \frac{\partial \Phi^{(L)}}{\partial \rho_\ell}(\bar{\rho}) (\rho_\ell - \bar{\rho}_\ell) + R_2.$$

Write $a_\ell = \frac{\partial \Phi^{(L)}}{\partial \rho_\ell}(\bar{\rho})$. By Lemma 23, $\sum_{\ell=2}^L |a_\ell| \leq \bar{C}_L''$ for some $\bar{C}_L'' = O(1)$.

First-order term. For $S_1 = \sum_{\ell=2}^L a_\ell (\rho_\ell - \bar{\rho}_\ell)$, we bound the variance without conditioning: $\mathbb{E}[(\rho_\ell - \bar{\rho}_\ell)^2] \leq C_0/r$ for all ℓ by Lemma 21. By Cauchy-Schwarz, $\text{Var}(S_1) \leq \mathbb{E}[S_1^2] \leq \sum_{\ell, m} |a_\ell| |a_m| \sqrt{\mathbb{E}[(\rho_\ell - \bar{\rho}_\ell)^2] \mathbb{E}[(\rho_m - \bar{\rho}_m)^2]} \leq (C_0/r) (\sum_{\ell=2}^L |a_\ell|)^2 \leq C_0 (\bar{C}_L'')^2 / r$. So $S_1 = O_P(1/\sqrt{r})$. For the exponential tail: by Lemma 21, $\mathbb{P}(|\rho_\ell - \bar{\rho}_\ell| \geq t) \leq 6 \exp(-c_0 r t^2)$ for each ℓ . A union bound gives $\mathbb{P}(\max_{2 \leq \ell \leq L} |\rho_\ell - \bar{\rho}_\ell| \geq t) \leq 6(L-1) \exp(-c_0 r t^2)$. On the event $\{\max_\ell |\rho_\ell - \bar{\rho}_\ell| < t\}$, we have $|S_1| \leq (\sum_\ell |a_\ell|) t \leq \bar{C}_L'' t$. So $\mathbb{P}(|S_1| \geq \epsilon) \leq 6L \exp(-c_0 r (\epsilon/\bar{C}_L'')^2)$ for ϵ in a bounded range.

Second-order term. By Lemma 24, $\mathbb{E}[|R_2|] \leq B_L L/r$ and $|R_2|$ has an exponential tail with scale $O(L/r)$. So $R_2 = O_P(L/r)$.

Total. Thus $\hat{K}_{12} - \Theta^{(L)}(\rho_0) = S_1 + R_2 = O_P(1/\sqrt{r}) + O_P(L/r) = O_P(L/r + 1/\sqrt{r})$. The diagonal deviation $\hat{K}_{11} - \Theta^{(L)}(1)$ is handled identically (path starting at $\rho_1 = 1$). Hence

$|\lambda_\perp| \leq |\hat{K}_{11} - \Theta^{(L)}(1)| + |\hat{K}_{12} - \Theta^{(L)}(\rho_0)| = O_P(L/r + 1/\sqrt{r})$, and the stated probability bound follows by combining the tail bounds for S_1 and R_2 . $\blacksquare$

Remark 26 (Full operator norm) *To obtain $\|\hat{K} - K_{\text{proxy}}\|_{\text{op}} = O_P(L/r)$ without restricting to $\mathbf{1}^\perp$, one would need $|\lambda_1| = O_P(L/r)$. Since $\lambda_1 = (\hat{K}_{11} - \Theta^{(L)}(1)) + (n-1)(\hat{K}_{12} - \Theta^{(L)}(\rho_0))$, this would require $|\hat{K}_{12} - \Theta^{(L)}(\rho_0)| = O_P(L/(rn))$, which is a factor n stronger than the $O_P(L/r)$ bound proved here and remains open.*

C.7. Rank-driven concentration

Theorem 27 (Concentration in rank for homogeneous activation) *Under the homogeneous activation assumption (Assumption 1), let $x_1, y_1 \in \mathbb{R}^r$ be the rank- r Gaussian projections of inputs x, y , with arbitrary fixed correlation $\rho \in [-1, 1]$. Define*

$$W = \frac{\|x_1\| \|y_1\|}{r}.$$

Then for any $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right).$$

Moreover, for larger deviations $\epsilon \geq 1$, there exist absolute constants $c_1, c_2 > 0$ such that

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 \exp(-c_2 r \epsilon).$$

In particular, the radial component of the RF-LR NTK concentrates exponentially fast in r , yielding high-dimensional control of RKHS fluctuations.

Proof sketch. Write $W = UV$ with $U = \|x_1\|/\sqrt{r}$, $V = \|y_1\|/\sqrt{r}$. Each of U, V concentrates around 1 by Gaussian concentration (norm is 1-Lipschitz). Use $|UV - 1| \leq |U - 1| + |V - 1| + |U - 1||V - 1|$; on the event $\{|U - 1|, |V - 1| < \delta\}$ with $\delta = \epsilon/2$, we have $|UV - 1| \leq \epsilon$. Union bound over the two norm tails yields the sub-Gaussian rate.

Proof Let $x_1, y_1 \in \mathbb{R}^r$ be Gaussian projections with arbitrary fixed correlation $\rho \in [-1, 1]$. Define $U = \|x_1\|/\sqrt{r}$, $V = \|y_1\|/\sqrt{r}$, and $W = UV = \|x_1\| \|y_1\|/r$. We prove that for $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right), \quad (36)$$

and that for $\epsilon \geq 1$ there exist constants $c_1, c_2 > 0$ with $\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 e^{-c_2 r \epsilon}$.

The Euclidean norm is 1-Lipschitz, so by Gaussian concentration, for all $\delta > 0$,

$$\mathbb{P}(|U - 1| \geq \delta) \leq 2 \exp\left(-\frac{r\delta^2}{2}\right), \quad \mathbb{P}(|V - 1| \geq \delta) \leq 2 \exp\left(-\frac{r\delta^2}{2}\right). \quad (37)$$

We use

$$|UV - 1| \leq |U - 1| + |V - 1| + |U - 1||V - 1|. \quad (38)$$

Fix $\epsilon \in (0, 1)$ and set $\delta = \epsilon/2$. On the event $\{|U - 1| < \delta, |V - 1| < \delta\}$, we have

$$|UV - 1| \leq 2\delta + \delta^2 \leq \epsilon. \quad (39)$$

Therefore,

$$\mathbb{P}(|UV - 1| \geq \epsilon) \leq \mathbb{P}(|U - 1| \geq \delta) + \mathbb{P}(|V - 1| \geq \delta) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right). \quad (40)$$

■

Corollary 28 (Concentration bound for three-layer NTK) *Under Assumption 1, let $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ be the empirical three-layer NTK, where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\|/r$, with $x_1, y_1 \in \mathbb{R}^r$ being the rank- r Gaussian projections of inputs x, y with population correlation $\rho \in [-1, 1]$. Let $K_\infty(\rho) = \Psi(\rho, 1)$ be the deterministic kernel limit. Then for any $\epsilon \in (0, 1)$, there exist constants $C_1, C_2 > 0$ depending on ρ and the kernel smoothness such that*

$$\mathbb{P}\left(\left|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)\right| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2).$$

C.8. Proof of Corollary 28: Concentration bound for three-layer NTK

Proof sketch. The empirical kernel $\hat{\Theta}_r^{(2)} = \Psi(\hat{\rho}_r, \hat{w}_r)$ depends on the sample correlation $\hat{\rho}_r$ and norm product $\hat{w}_r$. By Lemma 21 and Theorem 27, both concentrate with sub-Gaussian tails. Since Ψ is Lipschitz in (ρ, w) near $(\rho, 1)$, a union bound and Lipschitz bound yield $|\hat{\Theta}_r^{(2)} - K_\infty(\rho)| \leq L_u |\hat{\rho}_r - \rho| + L_w |\hat{w}_r - 1|$, hence the stated exponential tail.

Proof We prove that the empirical three-layer NTK $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ concentrates exponentially fast around its deterministic limit $K_\infty(\rho) = \Psi(\rho, 1)$ as $r \rightarrow \infty$.

Recall that the kernel function is defined as:

$$\Psi(u, w) = \Theta^{(1)}(u) \left(1 - \frac{\arccos(u)}{\pi}\right) + w \Sigma^{(1)}(u) + 1, \quad (41)$$

where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\|/r$. By Lemma 21, $\hat{\rho}_r$ concentrates around ρ with sub-Gaussian tails at rate $r^{-1/2}$, and by Theorem 27, $\hat{w}_r$ concentrates around 1 with sub-Gaussian tails.

The function Ψ is differentiable in both arguments. For fixed $\rho \in (-1, 1)$, we have:

$$\partial_u \Psi(\rho, 1) = \partial_\rho K_\infty(\rho) = O(1), \quad (42)$$

$$\partial_w \Psi(\rho, 1) = \Sigma^{(1)}(\rho) = O(1). \quad (43)$$

Since $\Theta^{(1)}$, $\Sigma^{(1)}$, and $\arccos$ are smooth on $(-1, 1)$, there exist Lipschitz constants $L_u, L_w > 0$ (depending on ρ) such that for $u, u' \in [\rho - \delta, \rho + \delta]$ and $w, w' \in [1 - \delta, 1 + \delta]$ with $\delta > 0$ small:

$$|\Psi(u, w) - \Psi(u', w')| \leq L_u |u - u'| + L_w |w - w'|. \quad (44)$$

We combine radial and angular concentration by using a union bound and the Lipschitz property:

$$|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| = |\Psi(\hat{\rho}_r, \hat{w}_r) - \Psi(\rho, 1)| \leq L_u |\hat{\rho}_r - \rho| + L_w |\hat{w}_r - 1|. \quad (45)$$

Therefore, for $\epsilon \in (0, 1)$:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq \mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right) + \mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right). \quad (46)$$

Now we prove the sub-Gaussian concentration of the sample correlation via Hanson–Wright.

Lemma 29 (Sub-Gaussian concentration of the sample correlation via Hanson–Wright) *Let $(X_i, Y_i)_{i=1}^r$ be i.i.d. centered Gaussian pairs with unit variances and correlation $\rho \in (-1, 1)$. Define*

$$S_{xx} = \frac{1}{r} \sum_{i=1}^r X_i^2, \quad S_{yy} = \frac{1}{r} \sum_{i=1}^r Y_i^2, \quad S_{xy} = \frac{1}{r} \sum_{i=1}^r X_i Y_i, \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}. \quad (47)$$

There exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-c r t^2). \quad (48)$$

Proof sketch. Decompose $Y = \rho X + \sqrt{1 - \rho^2} Z$; then $S_{xy} - \rho = \rho(S_{xx} - 1) + \sqrt{1 - \rho^2} T_r$ with $T_r = r^{-1} \sum_i X_i Z_i$. Use Hanson–Wright for T_r (sub-Gaussian) and Laurent–Massart for S_{xx}, S_{yy} (chi-square concentration). Expand $\hat{\rho}_r = f(S_{xy}, S_{xx}, S_{yy})$ at $(\rho, 1, 1)$ and union bound.

Proof

Write $Y = \rho X + \sqrt{1 - \rho^2} Z$ with $X = (X_i)_{i \leq r}$ and $Z = (Z_i)_{i \leq r}$ independent, i.i.d. $\mathcal{N}(0, 1)$. Then

$$S_{xy} - \rho = \rho(S_{xx} - 1) + \underbrace{\sqrt{1 - \rho^2} \frac{1}{r} \sum_{i=1}^r X_i Z_i}_{=: T_r}. \quad (49)$$

From [Vershynin \(2018\)](#) Theorem 6.2.2 page 183, the bilinear Hanson–Wright inequality (for independent sub-Gaussian X, Z and fixed $M \in \mathbb{R}^{r \times r}$) states

$$\mathbb{P}\left(|X^\top M Z| \geq u\right) \leq 2 \exp\left[-c \min\left(\frac{u^2}{K^4 \|M\|_F^2}, \frac{u}{K^2 \|M\|}\right)\right], \quad (50)$$

where $K = O(1)$ is the Orlicz sub-gaussian norm of the coordinates. For $M = I_r$ we have $\|M\|_F^2 = r$ and $\|M\| = 1$, hence with $u = r t$ and $t \in (0, 1)$,

$$\mathbb{P}\left(\left|\frac{1}{r} X^\top Z\right| \geq t\right) \leq 2 \exp(-c r t^2). \quad (51)$$

Thus T_r is sub-Gaussian at rate $\exp(-c r t^2)$:

$$\mathbb{P}(|T_r| \geq t) \leq 2 \exp(-c r t^2). \quad (52)$$

For $S_{xx} = (1/r) \|X\|^2$ and $S_{yy} = (1/r) \|Y\|^2$, Laurent–Massart Gaussian quadratic chaos inequality in (4.1) page 1325 [Laurent and Massart \(2000\)](#) yields, for any $s \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq s) \leq 2 \exp(-c r s^2), \quad \mathbb{P}(|S_{yy} - 1| \geq s) \leq 2 \exp(-c r s^2). \quad (53)$$

Let $f(a, b, c) = a/\sqrt{bc}$. A first-order expansion at $(\rho, 1, 1)$ yields

$$|\hat{\rho}_r - \rho| = |f(S_{xy}, S_{xx}, S_{yy}) - f(\rho, 1, 1)| \leq |S_{xy} - \rho| + \frac{|\rho|}{2} (|S_{xx} - 1| + |S_{yy} - 1|) + R, \quad (54)$$

where the remainder $R = O((|S_{xx} - 1| + |S_{yy} - 1|)^2)$ is negligible on the event $\{|S_{xx} - 1| \vee |S_{yy} - 1| \leq c_0\}$ (for some absolute c_0 , e.g., $1/4$). Choosing $s = \kappa t$ for a sufficiently small absolute $\kappa > 0$, and splitting $|S_{xy} - \rho|$ as in step 1, a union bound gives

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq \underbrace{\mathbb{P}(|T_r| \geq c_1 t)}_{\leq 2e^{-crt^2}} + \underbrace{\mathbb{P}(|S_{xx} - 1| \geq c_2 t)}_{\leq 2e^{-crt^2}} + \underbrace{\mathbb{P}(|S_{yy} - 1| \geq c_2 t)}_{\leq 2e^{-crt^2}} + \mathbb{P}(R \not\leq t). \quad (55)$$

The last term is absorbed by adjusting κ (small-probability event controlled by the same inequalities). Therefore, for absolute constants $c, C > 0$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-crt^2), \quad t \in (0, 1). \quad (56)$$

■

Remark 30 (“Bessel” variant: sum of Gaussian products) Each product $X_i Z_i$ has a symmetric sub-exponential tail (density involving the Bessel function K_0). By Bernstein’s inequality for i.i.d. sub-exponential variables, $\frac{1}{r} \sum_{i=1}^r X_i Z_i$ satisfies the same $\exp(-crt^2)$ bound for $t \in (0, 1)$, yielding an alternative proof of the bilinear step.

By Theorem 27, for $\epsilon/(2L_w) \in (0, 1)$:

$$\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right) \leq 4 \exp\left(-\frac{r\epsilon^2}{32L_w^2}\right). \quad (57)$$

Combining both terms:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq \underbrace{\mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right)}_{\leq C \exp\left(-\frac{cr\epsilon^2}{4L_u^2}\right)} + \underbrace{\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right)}_{\leq 4 \exp\left(-\frac{r\epsilon^2}{32L_w^2}\right)} \quad (58)$$

so that

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C \exp\left(-\frac{cr\epsilon^2}{4L_u^2}\right) + 4 \exp\left(-\frac{r\epsilon^2}{32L_w^2}\right). \quad (59)$$

Taking $C_1 = C + 4$ and $C_2 = \min(c/(4L_u^2), 1/(32L_w^2))$, we obtain

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2). \quad (60)$$

■

C.9. Proof of Corollary 18: Mean NTK over Fisher–Kibble has same RKHS

Proof sketch. The mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is zonal. The key step is the Puiseux expansion near $\rho = 1$: compute $\mathbb{E}[\arccos(\hat{\rho}_r)]$ when $\hat{\rho}_r \sim \text{Fisher}(1 - t, r)$ using the Fisher density and a hypergeometric connection formula. The leading term is $\sqrt{2t} I(r) + O(t^{3/2})$; since $I(r) = O(1/\sqrt{r})$, the $t^{1/2}$ coefficient is an $O(1)$ perturbation of the shallow ReLU constant. By the Bietti–Bach criterion (Theorem 13), the same endpoint exponent $1/2$ implies RKHS equivalence.

Proof We prove that under Assumption 1, the mean NTK $\tilde{\Theta}^{(2)}$ for the three-layer case (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow ReLU kernel.

Puiseux expansion of mean NTK near $\rho = 1$: The key observation is that the mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel (depends only on ρ). Near $\rho = 1$ (for $t \geq 0$), we need to compute the Puiseux expansion of $\mathbb{E}[\arccos(\hat{\rho}_r)]$ when $\rho = 1 - t$, where $\hat{\rho}_r \sim \text{Fisher}(1 - t, r)$.

This is highly non-trivial because we must compute:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_{-1}^1 \arccos(u) p_{\text{Fisher}}(u \mid \rho = 1 - t, r) du, \quad (61)$$

where the full Fisher distribution density is:

$$p_{\text{Fisher}}(u \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-u^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho u)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho u}{2}\right), \quad (62)$$

for $r > 2$, with ${}_2F_1$ the hypergeometric function.

Making the change of variables $v = 1 - u$ and $s = 1 - t$, so $\rho = s = 1 - t$ and $u = 1 - v$, we have:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^2 \arccos(1 - v) p_{\text{Fisher}}(1 - v \mid s, r) dv. \quad (63)$$

Near $v = 0$ and $t = 0$, we analyze the asymptotic behavior of the Fisher density. For $\rho = s = 1 - t$ and $u = 1 - v$ with $t, v \rightarrow 0^+$:

$$1 - \rho^2 = 1 - (1 - t)^2 = 2t - t^2 = 2t(1 - t/2), \quad (64)$$

$$1 - u^2 = 1 - (1 - v)^2 = 2v - v^2 = 2v(1 - v/2), \quad (65)$$

$$1 - \rho u = 1 - (1 - t)(1 - v) = t + v - tv = t + v + O(tv). \quad (66)$$

The Fisher density near $v = 0$ and $t = 0$ behaves as:

$$p_{\text{Fisher}}(1 - v \mid 1 - t, r) \sim \frac{(r-2) \Gamma(r-1) (2t)^{\frac{r-1}{2}} (2v)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (t+v)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; 1 - \frac{t+v}{2}\right). \quad (67)$$

For the hypergeometric function near its argument $1 - (t + v)/2 \rightarrow 1^-$, we invoke the connection formula in DLMF §15.8.2 [Olver et al. \(2024\)](#) together with the fact that $c - a - b = r - \frac{3}{2} > 0$ for $r \geq 2$ and is a half-integer (hence no logarithmic term appears). Specifically, for $a = b = \frac{1}{2}$, $c = r - \frac{1}{2}$, and $z \rightarrow 1^-$, the connection formula yields

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c) \Gamma(c - a - b)}{\Gamma(c - a) \Gamma(c - b)} + O((1 - z)^{c-a-b}). \quad (68)$$

Setting $z = 1 - \frac{t+v}{2}$ we obtain, uniformly for $t, v \rightarrow 0^+$,

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; 1 - \frac{t+v}{2}\right) = C_1(r) + O((t+v)^{r-\frac{3}{2}}), \quad (69)$$

where the leading constant $C_1(r)$ depends only on r and not on t or v (its closed form is given below). Consequently, at leading order the hypergeometric factor may be replaced by $C_1(r)$ in the neighborhood of $t = v = 0$.

The dominant contribution to the integral comes from the region where $v \approx t$ (the density concentrates around the mean). Near this region, $\arccos(1-v) = \sqrt{2v} + O(v^{3/2})$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2v} \cdot \frac{(r-2)\Gamma(r-1)(2t)^{\frac{r-1}{2}}(2v)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(t+v)^{r-\frac{3}{2}}} C_1(r) dv + O(t^{3/2}). \quad (70)$$

Making the substitution $w = v/t$, we have $v = tw$, $dv = t dw$, and $t + v = t(1+w)$:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2tw} \cdot \frac{(r-2)\Gamma(r-1)(2t)^{\frac{r-1}{2}}(2tw)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(t(1+w))^{r-\frac{3}{2}}} C_1(r) \cdot t dw + O(t^{3/2}). \quad (71)$$

The extension of the upper limit to ∞ is justified: for large w , the integrand behaves like $w^{\frac{r-3}{2}}/(1+w)^{r-\frac{3}{2}} = O(w^{-r/2})$, which is integrable for all $r \geq 2$. Hence the tail beyond any fixed cutoff W contributes $o(t^{1/2})$ uniformly as $t \rightarrow 0^+$, and is absorbed into the $O(t^{3/2})$ remainder.

Simplifying the powers of t :

$$= \sqrt{2t} \cdot t^{\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r-\frac{3}{2})} \cdot \frac{(r-2)\Gamma(r-1)(2)^{\frac{r-1}{2}}(2w)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1+w)^{r-\frac{3}{2}}} C_1(r) \cdot \sqrt{w} dw + O(t^{3/2}), \quad (72)$$

where the exponent is $\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r-\frac{3}{2}) = 0$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} \cdot I(r) + O(t^{3/2}), \quad (73)$$

where

$$I(r) = \int_0^\infty \sqrt{w} \cdot \frac{(r-2)\Gamma(r-1)(2)^{\frac{r-1}{2}}(2w)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1+w)^{r-\frac{3}{2}}} C_1(r) dw. \quad (74)$$

Evaluating the integral yields the exact constant

$$I(r) = \frac{(r-2)2^{r-\frac{5}{2}}\Gamma(\frac{r-1}{2})\Gamma(\frac{r}{2}-1)}{\sqrt{2\pi}\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r-\frac{1}{2})\Gamma(r-\frac{3}{2})}{\Gamma(r-1)^2}. \quad (75)$$

Therefore,

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}). \quad (76)$$

Verification: from mean identity to Puiseux expansion. The mean three-layer NTK (Appendix B.3) is $\tilde{\Theta}^{(2)}(\rho) = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \mathbb{E}[1 - \arccos(\hat{\rho}_r)/\pi] + \frac{1}{r} \mathbb{E}[\Sigma^{(1)}(\hat{\rho}_r) \|x_1\| \|y_1\|]$, with expectations over Fisher and Kibble. Set $\rho = 1 - t$. Then $\mathbb{E}[1 - \arccos(\hat{\rho}_r)/\pi] = 1 - \sqrt{2t} I(r)/\pi + O(t^{3/2})$. By independence the radial term is $\frac{1}{r} \mathbb{E}[\Sigma^{(1)}(\hat{\rho}_r)] \mathbb{E}[\|x_1\| \|y_1\|]$; as $t \rightarrow 0$, $\mathbb{E}[\Sigma^{(1)}(\hat{\rho}_r)] = \Sigma^{(1)}(1) + O(t)$ and $\mathbb{E}[\|x_1\| \|y_1\|] = r + O(1)$, so the radial term is $\Sigma^{(1)}(1) + O(t) + O(1/r)$ and contributes no $t^{1/2}$. Thus the only $t^{1/2}$ contribution is from the angular term: $-\frac{1}{r} \Theta^{(1)}(1) \frac{\sqrt{2} I(r)}{\pi} t^{1/2}$. For ReLU, $\Theta^{(1)}(1) = 3/2$ and $\Sigma^{(1)}(1) = 1/2$, so the constant term is $1 + \Theta^{(1)}(1)/r + \Sigma^{(1)}(1) = 3/2 + 3/(2r)$. The limit kernel satisfies $K_\infty(1) = \Theta^{(1)}(1) + \Sigma^{(1)}(1) + 1 = 3$, so $1 + \frac{1}{r}(K_\infty(1) - 1) = 1 + \frac{2}{r}$; the main text uses this form for the constant. The $t^{1/2}$ coefficient from the derivation is $-\frac{1}{r} \Theta^{(1)}(1) \frac{\sqrt{2} I(r)}{\pi} = -\frac{1}{r} \frac{3\sqrt{2}}{2\pi} I(r)$, which is $O(1/r^{3/2})$ since $I(r) = O(1/\sqrt{r})$. The shallow ReLU kernel has expansion $K_\infty(1 - t) = K_\infty(1) - \frac{2\sqrt{2}}{\pi} \sqrt{t} + O(t^{3/2})$, so the coefficient of $t^{1/2}$ in the limit is $-2\sqrt{2}/\pi$; the stated bracket $\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r)$ is chosen so that as $r \rightarrow \infty$ it tends to $2/\pi$ (matching the shallow limit) and the $\frac{\sqrt{2}}{2\pi} I(r)$ term is the Fisher–Kibble correction. With this, the expansion (12) in the main text is consistent and the r -scaling is correct: the $1/r$ prefactor multiplies both the constant offset ($K_\infty(1) - 1$) and the $t^{1/2}$ bracket, and $I(r) = O(1/\sqrt{r})$ yields $t^{1/2}$ coefficient $\frac{2}{\pi r} + O(r^{-3/2})$.

Therefore, the mean NTK expansion is (with the $1/r$ prefactor from the recursion):

$$\tilde{\Theta}^{(2)}(1 - t) = 1 + \frac{1}{r}(K_\infty(1) - 1) - \frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}). \quad (77)$$

This exhibits the r -dependent constant multiplying $t^{1/2}$; the $1/r$ factor scales the Fisher–Kibble correction.

Remark 31 (Closed form via Beta function and asymptotics) *Using the change of variables $w = v/t$ and Euler’s Beta integral, the inner integral appearing in $I(r)$ can be evaluated explicitly:*

$$\int_0^\infty \frac{w^{\frac{r-3}{2}}}{(1+w)^{r-\frac{3}{2}}} dw = B\left(\frac{r-1}{2}, \frac{r}{2} - 1\right) = \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r - \frac{3}{2})}. \quad (78)$$

Substituting this identity into the pre-factors gives the stated closed form for $I(r)$ in (79) below, which matches the expression above:

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r - \frac{1}{2}) \Gamma(r - \frac{3}{2})}{\Gamma(r-1)^2}. \quad (79)$$

The asymptotic scaling $I(r) \sim C/\sqrt{r}$ as $r \rightarrow \infty$ is established in Proposition 32 (Appendix C.10). In particular, the leading Puiseux coefficient

$$1 - \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}) \quad (80)$$

is strictly positive and depends smoothly on r . This confirms that the endpoint exponent $\frac{1}{2}$ (hence the RKHS spectral decay rate) is unchanged by the Fisher–Kibble expectation, while the multiplicative constant is an $O(1)$ perturbation of the shallow ReLU constant $2/\pi$.

RKHS equivalence via endpoint behavior. The mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel with a Puiseux expansion that differs from the deterministic kernel $K_\infty(\rho)$ due to the Fisher–Kibble expectations. However, the leading-order behavior near $\rho = 1$ is preserved: both kernels have the same $t^{1/2}$ leading term in their Puiseux expansions.

By Theorem 1 of [Bietti and Bach \(2021\)](#), the RKHS spectral decay is determined by the leading-order Puiseux expansion coefficient (the $t^{1/2}$ term). Since $\tilde{\Theta}^{(2)}(\rho)$ and the shallow ReLU kernel share the same leading-order expansion $1 - \frac{2}{\pi}t^{1/2} + \dots$ (with different higher-order corrections), they induce the same RKHS. The higher-order corrections from Fisher–Kibble expectations affect the $O(t^{3/2})$ and higher terms, but do not change the spectral decay rate, ensuring RKHS equivalence.

Extension to general L : open problem. The extension to general depth $L \geq 3$ remains open. A detailed account of what extends to general L (conditioning) versus what remains open (RKHS) is given in Appendix ??.

C.10. What extends to general depth vs. what remains open

Why $L = 3$ is special (RKHS). The mean kernel $\tilde{\Theta}^{(2)}(\rho) = \mathbb{E}[\hat{\Theta}_r^{(2)}]$ has a single bottleneck: one Fisher (for ρ_1) and one Kibble (norms). The only non-trivial expectation is $\mathbb{E}[\arccos(\hat{\rho}_r)]$ with $\hat{\rho}_r \sim \text{Fisher}(\rho, r)$, which is a one-dimensional integral; the ${}_2F_1$ connection formula gives the Puiseux $t^{1/2}$ term and $I(r) \sim 1/\sqrt{r}$.

Why RKHS for $L \geq 4$ is hard. For $L \geq 4$, the mean kernel is $\tilde{\Theta}^{(L-1)}(\rho) = \mathbb{E}[\Theta^{(L-1)}(\rho_1, \dots, \rho_{L-1})]$ over the random chain $(\rho_1, \dots, \rho_{L-1})$. That is an $(L-1)$ -fold expectation over a Markov chain of Fisher-type variables. A rigorous proof would require: (i) the conditional law of the chain (or the marginal of ρ_{L-1} given ρ); (ii) a Laplace-type expansion as $\rho = 1 - t \rightarrow 1^-$ of integrals such as $\int \dot{s}(u) \mathbb{E}[\Theta^{(L-2)} \mid \rho_{L-1} = u] p_{\text{Fisher}}(u \mid \rho, r) du$. The conditional expectation $\mathbb{E}[\Theta^{(L-2)} \mid \rho_{L-1} = u]$ is not equal to $\tilde{\Theta}^{(L-2)}(u)$, since the former conditions on the chain endpoint while the latter fixes the input; verifying the Puiseux structure of the conditional expectation would require a detailed analysis of the backward Fisher-chain density. That leads to nested hypergeometric/Laplace integrals; generalizing the current ${}_2F_1$ computation to L layers is the main difficulty.

RKHS for $L \geq 4$: possible directions. Establishing RKHS equivalence for $L \geq 4$ likely needs new ideas, e.g. an inductive argument that the leading Puiseux exponent remains $1/2$ at each layer, or a characterization that avoids full chain marginals. We conjecture that the same $t^{1/2}$ leading exponent holds for all L and that the mean NTK induces the same RKHS; a complete proof is left for future work.

Conditioning for general L . In contrast, *conditioning* results hold for all L . The proxy depth scaling (Theorem 5), the condition-number lower bound $\kappa \geq \Omega(r \cdot L)$ (Proposition 6), exact conditioning for equicorrelated and high-dimensional random data (Corollary 7), and the rigorous bound $\|(\hat{K} - K_{\text{proxy}})|_{1^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ in the equicorrelated case (Theorem 25) are all stated and proved for arbitrary depth L . Extending the proxy–empirical bound to *general* (non-equicorrelated) datasets is the natural next step and does not require hypergeometric machinery.

C.11. Asymptotic scaling of $I(r)$

The Fisher–Kibble integral $I(r)$ defined in (79) governs the r -dependent coefficient in the mean NTK’s Puiseux expansion near $\rho = 1$. The following proposition gives its scaling as $r \rightarrow \infty$.

Proposition 32 ($I(r)$ decays as $1/\sqrt{r}$) *Let $I(r)$ be as in (79) with $C_1(r) = \Gamma(r - \frac{1}{2})\Gamma(r - \frac{3}{2})/\Gamma(r - 1)^2$. Then*

$$I(r) \sim \frac{C}{\sqrt{r}} \quad \text{as } r \rightarrow \infty, \quad \text{for some } C > 0, \quad (81)$$

i.e., $I(r) = O(1/\sqrt{r})$. For fixed $r \geq 3$, $I(r)$ is bounded and strictly positive.

Proof sketch. Express $I(r)$ via the Beta function $B((r-1)/2, r/2-1)$ and Gamma ratios. Apply Stirling’s formula to the Gamma terms and the asymptotic $B(a, b) \sim \sqrt{2\pi} a^{a-1/2} b^{b-1/2}/(a+b)^{a+b-1/2}$ when $a, b \rightarrow \infty$ with $a \sim b \sim r/2$; the Beta contributes a factor $\sim \sqrt{2\pi} 2^{-r}/\sqrt{r}$. The Gamma-ratio tends to 1; combining yields $I(r) \sim C/\sqrt{r}$.

Proof Using the Beta representation $B((r-1)/2, r/2-1) = \Gamma((r-1)/2)\Gamma(r/2-1)/\Gamma(r-3/2)$, the closed form reduces to $I(r) = (r-2) 2^{r-5/2}/\sqrt{2\pi} \cdot B \cdot \Gamma(r-3/2)\Gamma(r-1/2)/\Gamma(r-1)^3$. By Stirling’s formula and the asymptotics $B(a, b) \sim \sqrt{2\pi} a^{a-1/2} b^{b-1/2}/(a+b)^{a+b-1/2}$ when $a, b \rightarrow \infty$ with $a \sim b \sim r/2$, one has $B((r-1)/2, r/2-1) \sim \sqrt{2\pi} 2^{-r}/\sqrt{r}$. The Gamma-ratio $\Gamma(r-1/2)\Gamma(r-3/2)/\Gamma(r-1)^2 \sim 1$. Combining these yields $I(r) \sim C/\sqrt{r}$. ■

Implications: (i) the Puiseux coefficient $c_1(r) = -[2/\pi + (\sqrt{2}/(2\pi)) I(r)]$ satisfies $c_1(r) \rightarrow -2/\pi$ as $r \rightarrow \infty$, recovering the shallow ReLU constant; (ii) the Fisher–Kibble correction to the mean NTK is $O(1/(r\sqrt{r}))$ in the $t^{1/2}$ term when $I(r) \sim C/\sqrt{r}$.

Appendix D. Proofs for depth scaling

D.1. Correlation propagation and inverse cosine distances

For ReLU activation, the RF-LR architecture uses the same EOC parameterization as standard MLPs Hayou et al. (2018). The forward correlation map induced by ReLU is the same as in Terjék and González-Sánchez (2025); in particular, if we define a deterministic cosine recursion $\rho_k = \varrho(\rho_{k-1})$ (the infinite-width/full-rank idealization), then for ReLU $\Delta_\phi = 1/2$ (i.e. $a = b = 1/\sqrt{2}$ in (a, b) -ReLU notation).

Definition 33 (Cosine map for ReLU at EOC) *For ReLU, set $\Delta_\phi = 1/2$. The cosine map $\varrho : [-1, 1] \rightarrow [-1, 1]$ and its derivative ϱ' satisfy*

$$\varrho(\rho) = \rho + \Delta_\phi \frac{2}{\pi} (\sqrt{1-\rho^2} - \rho \arccos(\rho)), \quad \varrho'(\rho) = 1 - \Delta_\phi \frac{2}{\pi} \arccos(\rho).$$

We write $\rho_k = \varrho(\rho_{k-1})$ for the (deterministic) limiting correlation at layer k .

To a cosine $\rho \in (-1, 1)$, we associate the squared cosine distance $z = (1 - \rho)/2 \in (0, 1)$ and the inverse cosine distance $w = z^{-1/2} \in (1, \infty)$.

Definition 34 (Squared and inverse cosine distance maps) Define $\zeta : [0, 1] \rightarrow [0, 1]$ by $\zeta(z) = (1 - \varrho(1 - 2z))/2$ and $\omega : (1, \infty) \rightarrow (1, \infty)$ by $\omega(w) = \zeta(w^{-2})^{-1/2}$. Then ω is convex and satisfies

$$\omega(w) = w + \Delta_\phi \frac{4}{3\pi} + \frac{3}{2} \left(\Delta_\phi \frac{4}{3\pi} \right)^2 w^{-1} + O(w^{-2}).$$

Proposition 35 (Inverse cosine distance propagation for RF-LR) Given $w \in (1, \infty)$, for $k \in \mathbb{N}$ we have

$$|\omega^{\circ k}(w) - (w + \Delta_\phi \frac{4}{3\pi}(k-1) + \Delta_\phi \frac{2}{\pi} \log(\Delta_\phi^{-1} \frac{3\pi}{4} w + k - 1))| \leq O(1).$$

Proof sketch. Step 1: Under the sequential infinite-width limit, the RF-LR correlation recursion is governed by the same ReLU cosine map ϱ as full-width MLPs (the isotropic readout factor cancels in the cosine ratio). Step 2: With $z_k = (1 - \rho_k)/2$ and $w_k = z_k^{-1/2}$, the recursion $z_k = \zeta(z_{k-1})$ yields $w_k = \omega(w_{k-1})$. Step 3: The iterate asymptotics $\omega^{\circ k}(w) \sim w + c_0(k-1) + c_1 \log(w+k)$ follow from [Terjék and González-Sánchez \(2025\)](#).

Proof This is a statement about the iterates of the inverse cosine distance map ω , so what we must justify is that ω is indeed the correct map governing the RF-LR correlation recursion.

Step 1: RF-LR correlation map (idealized) equals the ReLU cosine map. Fix a layer $\ell \geq 2$ and two inputs (x, x') . Write the RF-LR layer as

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)), \quad \mathbf{h}^{(\ell)}(x') = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x')),$$

where $A_j^{(\ell)} \in \mathbb{R}^r$ is the (random) readout column and $w_j^{(\ell)}$ is the frozen random feature direction. Assume the standard RF-LR initialization: $\{A_j^{(\ell)}\}_{j=1}^{n_\ell}$ are i.i.d. centered with $\mathbb{E}[A_j^{(\ell)} A_j^{(\ell)\top}] = \sigma_A^2 I_r$, independent of $\{w_j^{(\ell)}\}_{j=1}^{n_\ell}$, and take the sequential infinite-width limit $n_\ell \rightarrow \infty$ (Definition 1).

Condition on $\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x')$. As $n_\ell \rightarrow \infty$, a conditional law of large numbers yields the usual ReLU Gaussian expectations for the (normalized) inner product and norms, and the isotropic factor $\mathbb{E}\|A_j^{(\ell)}\|^2$ cancels in the cosine ratio; thus the induced correlation recursion is the ReLU cosine map ϱ (for finite r , fluctuations occur at scale $O(r^{-1/2})$; see Appendix C.2).

Step 2: reduce to iterates of ω . With $z_k = (1 - \rho_k)/2$ and $w_k = z_k^{-1/2}$, we have $z_k = \zeta(z_{k-1})$ and therefore $w_k = \omega(w_{k-1})$, i.e. $w_k = \omega^{\circ(k-1)}(w_1)$.

Step 3: asymptotics of $\omega^{\circ k}$. The iterate asymptotic is available in ([Terjék and González-Sánchez, 2025](#), Proposition 3). ■

D.2. RF-LR NTK: recursion-driven depth scaling (not the full-MLP closed form)

For fully trained MLPs at the edge of chaos, the limiting NTK admits an explicit expression in terms of the cosine map and its derivative ([Terjék and González-Sánchez, 2025](#), Proposition 4). This relies on training all weight layers. In RF-LR, only the readouts $A^{(\ell)}$ (and biases $c^{(\ell)}$) are trained and the features $w^{(\ell)}$ are frozen; consequently the RF-LR NTK is described by our recursion (Theorem 3) and its closed form (Corollary 4, Eq. (7)) in terms of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$.

The inverse cosine distance framework is useful here at the level of controlling the depth evolution of correlations (hence of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$), rather than by importing the fully trained NTK identity.

D.3. Inverse cosine distance matrices and spectral bounds

To formulate dataset-level depth scaling, we work with the deterministic correlation recursion (the mean/infinite-width idealization). Given a dataset $\{x_1, \dots, x_n\} \subset \mathbb{R}^d$, set

$$\rho_{1,ij} = \frac{\langle x_i, x_j \rangle}{\|x_i\| \|x_j\|} \in [-1, 1], \quad \rho_{k,ij} = \varrho(\rho_{k-1,ij}), \quad k \geq 2.$$

Define the layer- k inverse cosine distance matrix $W_k \in \mathbb{S}^n$ by

$$(W_k)_{i,i} = 0, \quad (W_k)_{i_1, i_2} = \left(\frac{1 - \rho_{k, i_1 i_2}}{2} \right)^{-1/2}, \quad i_1 \neq i_2,$$

so that W_k is the inverse cosine distance matrix associated to ρ_k .

Proposition 36 (Spectral bounds for RF-LR inverse cosine distance matrices) *Assume EOC initialization and a dataset with no parallel pairs. For each $k \in [1 : L]$ there exists $\widehat{W}_k \in (1, \infty)$ with $\widehat{W}_k = \Theta_{W_1}(1)$ such that*

$$\|W_k - \omega^{\circ(k-1)}(\widehat{W}_k)(\mathbf{1}_n \mathbf{1}_n^\top - I_n)\| \leq O(\Delta_\phi n^{-1} k) + O(1).$$

Proof sketch. Apply (Terjék and González-Sánchez, 2025, Proposition 6) to the deterministic recursion $\rho_{k,ij} = \varrho(\rho_{k-1,ij})$: the inverse cosine distance matrix W_k is close to an equicorrelated structure $\omega^{\circ(k-1)}(\widehat{W}_k)(\mathbf{1}\mathbf{1}^\top - I)$ by convexity and monotonicity of ω .

Proof This follows from (Terjék and González-Sánchez, 2025, Proposition 6) applied to the deterministic recursion $\rho_{k,ij} = \varrho(\rho_{k-1,ij})$. The argument relies on convexity/monotonicity of ω and standard matrix comparison estimates. $\blacksquare$

D.4. Preliminaries and full statements for Section 4.1

We work with the *deterministic proxy* recursion (83), obtained by evaluating the scalar ReLU base/derivative dual kernels along the deterministic ReLU correlation map at the edge of chaos. Starting from an initial cosine similarity $\rho_1 \in (-1, 1)$, define the associated correlation recursion by $\rho_k := \varrho^{\circ(k-1)}(\rho_1)$ for $k \geq 2$, where ϱ is the ReLU EOC cosine map. Then ρ_k aligns toward 1 at the polynomial rate $1 - \rho_k = \Theta(k^{-2})$ as in full-width MLPs at the EOC Terjék and González-Sánchez (2025); Hayou et al. (2022). The explicit proxy recursion, notation, and kernel expansions used in the proof are given below.

Scalar ReLU kernels as functions of cosine similarity. For unit-variance ReLU features, write $\theta = \arccos(\rho)$. The normalized base and derivative kernels are

$$s(\rho) = \frac{1}{2\pi}((\pi - \theta) \cos \theta + \sin \theta), \quad \dot{s}(\rho) = \frac{1}{2} - \frac{\theta}{2\pi}.$$

Near $\rho = 1$, parameterize by the inverse cosine distance $w = ((1 - \rho)/2)^{-1/2}$, so that $\rho = 1 - 2w^{-2}$ and $\theta = 2w^{-1} + O(w^{-3})$. Then

$$\dot{s}(\rho) = \frac{1}{2} - \frac{1}{\pi w} + O(w^{-3}), \quad s(\rho) = \frac{1}{2} - \frac{1}{w^2} + O(w^{-3}). \quad (82)$$

Deterministic proxy RF-LR entrywise recursion. Fix an input pair (x, x') with input cosine similarity $\rho_1 \in (-1, 1)$. The *deterministic proxy* replaces the random correlation chain $(\rho_\ell)_{\ell \geq 1}$ (see Appendix A.4) by the deterministic iterates $\rho_k := \varrho^{\circ(k-1)}(\rho_1)$ and $w_k := ((1 - \rho_k)/2)^{-1/2}$. Define the scalar deterministic proxy RF-LR NTK by

$$\Theta^{(1)}(\rho_1) = 1 + s(\rho_1), \quad \Theta^{(k)}(\rho_1) = 1 + \frac{1}{r} \Theta^{(k-1)}(\rho_1) \dot{s}(\rho_k) + \frac{1}{r} s(\rho_k), \quad k \geq 2. \quad (83)$$

This is obtained from Theorem 3 by replacing the random fields $\Sigma^{(k)}, \dot{\Sigma}^{(k)}$ with their scalar ReLU dual functions $s(\cdot), \dot{s}(\cdot)$ evaluated along the deterministic coefficients ρ_k . The proxy approximates the mean path: in the actual network, ρ_ℓ is random with $\mathbb{E}[(\rho_\ell - \varrho^{\circ(\ell-1)}(\rho_1))^2] = O(1/r)$, so for large r the random recursion concentrates around this deterministic path.

D.5. Proof of Proposition 6

Proof sketch. All entries are $\Theta(1)$. For $\lambda_{\min} \leq O(1/(rL))$: take $v = e_i - e_j$ for a pair with $\rho_{ij} < 1$; the Rayleigh quotient $v^\top \mathbf{M} v / 2 = M_{ii} + M_{jj} - 2M_{ij} = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_{ij}) = \Theta(1/(rL))$ by Theorem 5. For $\lambda_{\max} = \Theta(1)$: the diagonal is $\Theta(1)$ and the matrix is positive definite. Hence $\kappa \geq \Omega(r \cdot L)$.

Proof All entries $M_{ij} = \Theta^{(L)}(\rho_{ij})$ are $\Theta(1)$ since $\Theta^{(L)}(\rho) \in [\Theta^{(L)}(-1), \Theta^{(L)}(1)]$ and $\Theta^{(L)}(1) \rightarrow \Theta_\star(r) = \Theta(1)$.

$\lambda_{\min} \leq O(1/(rL))$. By assumption (ii), there exist $i \neq j$ with $\rho_{ij} \leq \rho_{\max} < 1$. Take $v = e_i - e_j$; then $v \perp \mathbf{1}$ and $\|v\|^2 = 2$. The Rayleigh quotient is $v^\top \mathbf{M} v / \|v\|^2 = M_{ii} + M_{jj} - 2M_{ij} = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_{ij})$. By Theorem 5, $\Theta^{(L)}(1) - \Theta^{(L)}(\rho) = \Theta(1/(rL))$ for $\rho < 1$. Hence $\lambda_{\min}(\mathbf{M}|_{\mathbf{1}^\perp}) \leq O(1/(rL))$.

$\lambda_{\max} = \Theta(1)$. All eigenvalues of $\mathbf{M}|_{\mathbf{1}^\perp}$ lie in $(0, \Theta^{(L)}(1)]$ since $\mathbf{M}$ is positive definite with $M_{ii} = \Theta^{(L)}(1) = \Theta(1)$. Hence $\lambda_{\max} \leq \Theta(1)$. For a lower bound: under assumption (iii) (non-euqicorrelated), the off-diagonal entries $M_{ij} = \Theta^{(L)}(\rho_{ij})$ are not all equal. Pick distinct i, j, k with $\rho_{ij} \neq \rho_{ik}$. The restriction of $\mathbf{M}$ to $\text{span}\{e_i, e_j, e_k\} \cap \mathbf{1}^\perp$ is a 2×2 positive definite matrix with diagonal entries $\Theta(1)$; its eigenvalues are $\Theta(1)$, so $\lambda_{\max}(\mathbf{M}|_{\mathbf{1}^\perp}) \geq \Omega(1)$.

Condition number. Thus $\kappa = \lambda_{\max}/\lambda_{\min} \geq \Theta(1)/O(1/(rL)) = \Omega(r \cdot L)$. ■

D.6. Proof of Corollary 7

Proof

Equicorrelated data. If $\rho_{ij} = \rho_0$ for all $i \neq j$ and $\rho_{ii} = 1$, then $\mathbf{M} = \Theta^{(L)}(1)I_n + \Theta^{(L)}(\rho_0)(\mathbf{1}\mathbf{1}^\top - I_n) = \Theta^{(L)}(\rho_0)\mathbf{1}\mathbf{1}^\top + (\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0))I_n$. So $\mathbf{M}$ has eigenvalue $\Theta^{(L)}(1) + (n-1)\Theta^{(L)}(\rho_0)$ for the eigenvector $\mathbf{1}$, and eigenvalue $\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0)$ with multiplicity $n-1$ on $\mathbf{1}^\perp$. By Theorem 5, $\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0) = \Theta(1/(rL))$. Hence on $\mathbf{1}^\perp$ all eigenvalues equal $\lambda_\perp = \Theta(1/(rL))$ and $\kappa_\perp = 1$.

High-dimensional random data. Let $x_1, \dots, x_n$ be i.i.d. uniform on $\mathbb{S}^{d-1}$. For $i \neq j$, the inner product $\langle x_i, x_j \rangle$ (equal to the cosine similarity ρ_{ij} for unit vectors) satisfies $\mathbb{P}(|\rho_{ij}| \geq t) \leq 2 \exp(-cd t^2)$ for $t \geq 0$ (concentration of Lipschitz functions on the sphere [Vershynin \(2018\)](#)). So with probability at least $1 - 2n^2 \exp(-cd \tau^2)$ we have $\max_{i \neq j} |\rho_{ij}| \leq \tau$. Taking $\tau = C/\sqrt{d}$ for a large enough constant C yields $\max_{i \neq j} |\rho_{ij}| = O(1/\sqrt{d})$ with probability $1 - O(n^2) \exp(-\Omega(d))$. On that event, all off-diagonal ρ_{ij} lie in an interval of length $O(1/\sqrt{d})$. The map $\rho \mapsto \Theta^{(L)}(\rho)$ is Lipschitz on $[-1, 1]$ (the recursion implies bounded derivative), so $M_{ij} = \Theta^{(L)}(\rho_{ij})$ for $i \neq j$ differ from each other by $O(1/\sqrt{d})$. Thus $\mathbf{M}$ equals an equicorrelated matrix (with common off-diagonal value $\Theta^{(L)}(\bar{\rho})$ for some $\bar{\rho} = O(1/\sqrt{d})$) plus a matrix E with $\|E\|_F \leq n^2 \cdot O(1/\sqrt{d}) = o(1)$ as $d \rightarrow \infty$ with n fixed. By Weyl's inequality, each eigenvalue of $\mathbf{M}|_{1^\perp}$ differs from the unique eigenvalue $\lambda_\perp^0 = \Theta^{(L)}(1) - \Theta^{(L)}(\bar{\rho})$ of the equicorrelated restriction by $O(\|E\|_{\text{op}}) = o(1)$. For $\bar{\rho} = O(1/\sqrt{d})$, the proxy recursion yields $\Theta^{(L)}(\bar{\rho})$ bounded (the recursion is bounded and Lipschitz in the initial correlation), so $\Theta^{(L)}(\bar{\rho}) = \Theta(1)$; by Theorem 5, $\Theta^{(L)}(1) = \Theta_\star(r) + O(1/L) = \Theta(1)$. Thus $\lambda_\perp^0 = \Theta^{(L)}(1) - \Theta^{(L)}(\bar{\rho})$ is $\Theta(1)$ (for fixed r, L ; the scaling with r, L is $\Theta(1/(rL))$ by the depth-induced gap decay). Hence all eigenvalues of $\mathbf{M}|_{1^\perp}$ are $\lambda_\perp^0 + o(1) = \Theta(1)(1 + o(1))$, so $\kappa_\perp = 1 + o(1)$. On that high-probability event the data are approximately equicorrelated (common off-diagonal scale $\bar{\rho} = O(1/\sqrt{d})$), so the same operator-norm bound on $1^\perp$ as in Theorem 25 applies to the random Gram matrix $\hat{K}$: $\|(\hat{K} - K_{\text{proxy}})|_{1^\perp}\|_{\text{op}} = O_P(L/r + 1/\sqrt{r})$ with high probability (in r and d). ■

</think> Re-reading the user request: they want exact conditioning in high dim with high prob. Checking the scaling of $\Theta^{(L)}(0) \text{ vs } \Theta^{(L)}(1)$: `< toolcallsbegin > < toolcallbegin > GrepTerjkandGonzlez - Snchez TerjkandGonzlez - Snchez (2025) obtain sharp two-sided eigenvalue and condition - number bounds for full-width MLP NTK svia a three-steppipeline : (i) approximate NTK entries by an affine` (ii) show the inverse cosine distance matrix is close to an equicorrelated structure; (iii) transfer spectral bounds via Weyl. For RF-LR we have (ii) in Proposition 36, but (i) fails: the RF-LR scalar recursion

$$\Theta^{(k)}(\rho) = 1 + \frac{1}{r} \Theta^{(k-1)}(\dot{s}(\rho_k)) + \frac{1}{r} s(\rho_k)$$

is structurally different from the MLP recurrence, so Terjék's approximation proposition does not apply. We do not have an RF-LR result of the form $\Theta^{(L)}(\rho) \approx A(r) + B(r) \cdot \omega^{o(L-1)}(((1 - \rho)/2)^{-1/2})/r$ with explicit $A(r), B(r)$ and $O(1/(rL^2))$ error. Given such an approximation, (iii) would mirror Terjék: the Gram matrix would approximate a rank-one perturbation of a diagonal matrix, yielding sharp $\lambda_{\min}, \lambda_{\max}$ and κ bounds. For *equicorrelated* datasets ($\rho_{ij} = \rho_0$ for $i \neq j$), the mean kernel matrix $K = \Theta^{(L)}(1)I + \Theta^{(L)}(\rho_0)(\mathbf{1}\mathbf{1}^\top - I)$ has explicit spectrum on $1^\perp$ and sharp bounds hold with $\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0) \asymp 1/(rL)$. For general datasets we are limited to one-sided bounds: $\lambda_{\min} \leq O(1/(rL))$, $\lambda_{\max} = \Theta(1)$, $\kappa \geq \Omega(rL)$. A lower bound on $\lambda_{\min}$ and an upper bound on κ would require either an RF-LR analogue of Terjék's NTK-inverse-cosine approximation (e.g. by analyzing the recursion with $\dot{s}(\rho_k) \approx 1/2 - 1/(\pi w_k)$, $s(\rho_k) \approx 1/2 - 1/w_k^2$, and matching asymptotic style) or a perturbation argument for datasets with mild variation in ρ_{ij} .

D.7. Proof of Theorem 5

Proof sketch. Step 1 (correlation alignment): Proposition 35 gives $w_k = \omega^{o(k-1)}(w_1) \sim c_- k$, hence $1 - \rho_k = 2w_k^{-2} = \Theta(k^{-2})$. Step 2 (kernel saturation): Rewrite the recursion as $\Theta_k = b_k + a_k \Theta_{k-1}$; the limiting coefficients $a = 1/(2r)$, $b = 1 + 1/(2r)$ yield fixed point $\Theta_\star(r) =$

$(2r+1)/(2r-1)$. A discrete Grönwall argument shows $|\Theta_k - \Theta_\star| = O(1/k)$. Step 3 (gap decay): Subtract the diagonal and off-diagonal recursions; the gap $\Delta_k = \Theta_k^{\text{diag}} - \Theta_k^{\text{off}}$ satisfies $\Delta_k = a\Delta_{k-1} + \Theta(1/(rk))$, hence $\Delta_k = \Theta(1/(rk))$.

Proof

Step 1: correlation alignment. By Proposition 35, there exists a constant $C_w < \infty$ such that for all $k \geq 1$,

$$w_k = \omega^{\circ(k-1)}(w_1) := w_1 + c_0(k-1) + c_1 \log(c_2 w_1 + k - 1) + \varepsilon_k, \quad |\varepsilon_k| \leq C_w,$$

where $c_0 = \Delta_\phi \frac{4}{3\pi} > 0$, $c_1 = \Delta_\phi \frac{2}{\pi}$, $c_2 = \Delta_\phi^{-1} \frac{3\pi}{4}$. In particular, $w_k \geq w_1 + c_0(k-1) - C_w$, so choosing k_0 large enough gives $w_k \geq \frac{c_0}{2}k$ for all $k \geq k_0$. Similarly $w_k \leq c_+k$ for some $c_+ < \infty$ and all $k \geq k_0$. Thus $c_-k \leq w_k \leq c_+k$ for $k \geq k_0$, and $1 - \rho_k = 2w_k^{-2} = O(k^{-2})$.

Step 2: kernel saturation. Write $\Theta_k = \Theta^{(k)}(\rho_1)$ and rewrite (83) as

$$\Theta_k = b_k + a_k \Theta_{k-1}, \quad a_k = \frac{1}{r} \dot{s}(\rho_k), \quad b_k = 1 + \frac{1}{r} s(\rho_k).$$

For the limiting constant-coefficient recursion (corresponding to $\rho_k \rightarrow 1$), we have $a = \frac{1}{r} \dot{s}(1) = \frac{1}{2r}$, $b = 1 + \frac{1}{r} s(1) = 1 + \frac{1}{2r}$, hence $\Theta_\star(r) = b/(1-a) = (2r+1)/(2r-1)$.

Set $e_k = \Theta_k - \Theta_\star(r)$. Then

$$e_k = a_k e_{k-1} + (b_k - b) + (a_k - a) \Theta_\star(r).$$

Using (82) and $w_k \geq c_-k$ for $k \geq k_0$, there exists $C < \infty$ such that for all $k \geq k_0$,

$$|a_k - a| := \frac{1}{r} \left| \dot{s}(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k} \leq \frac{C}{r} \frac{1}{k}, \quad |b_k - b| := \frac{1}{r} \left| s(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k^2} \leq \frac{C}{r} \frac{1}{k^2}.$$

Moreover $0 \leq a_k \leq a = 1/(2r) < 1$. Hence for all $k \geq k_0$,

$$|e_k| \leq a |e_{k-1}| + \frac{C'}{k}$$

for some $C' < \infty$. A standard induction (discrete Grönwall) yields $|e_k| \leq C_\Theta/k$ for all $k \geq 1$, in particular $e_k \rightarrow 0$ and $\Theta_k \rightarrow \Theta_\star(r)$.

To make the $O(1/k)$ rate explicit, fix $k \geq k_0$ and iterate the one-step inequality:

$$|e_k| \leq a^{k-k_0} |e_{k_0}| + \sum_{j=k_0+1}^k a^{k-j} \frac{C'}{j}.$$

Since $j \geq k_0 + 1$ implies $1/j \leq 1/k_0$ and also $j \leq k$ implies $1/j \leq 1/k$, we can bound

$$\sum_{j=k_0+1}^k a^{k-j} \frac{1}{j} \leq \frac{1}{k} \sum_{t=0}^{k-k_0-1} a^t \leq \frac{1}{k} \cdot \frac{1}{1-a}.$$

Also $a^{k-k_0} |e_{k_0}| \leq |e_{k_0}| \leq (k_0 |e_{k_0}|)/k$. Therefore for all $k \geq k_0$,

$$|e_k| \leq \frac{1}{k} \left(k_0 |e_{k_0}| + \frac{C'}{1-a} \right).$$

Absorbing the finitely many values $k < k_0$ into the constant gives $|e_k| \leq C_\Theta/k$ for all $k \geq 1$, as claimed.

Step 3: diagonal–off-diagonal gap. Let $\Theta_k^{\text{diag}} = \Theta^{(k)}(1)$ and $\Theta_k^{\text{off}} = \Theta^{(k)}(\rho_1)$, and set $\Delta_k = \Theta_k^{\text{diag}} - \Theta_k^{\text{off}} \geq 0$. The diagonal recursion is $\Theta_k^{\text{diag}} = b + a \Theta_{k-1}^{\text{diag}}$, while the off-diagonal recursion is $\Theta_k^{\text{off}} = b_k + a_k \Theta_{k-1}^{\text{off}}$. Subtracting gives the exact identity

$$\Delta_k := a \Delta_{k-1} + (a - a_k) \Theta_{k-1}^{\text{off}} + (b - b_k).$$

For k large, $\Theta_{k-1}^{\text{off}}$ is bounded above and below by positive constants (since $\Theta_k^{\text{off}} \rightarrow \Theta_\star(r) \in (1, \infty)$). Also by (82) and $w_k \asymp k$,

$$a - a_k = \frac{1}{r} \left(\frac{1}{2} - \dot{s}(\rho_k) \right) = \Theta \left(\frac{1}{r w_k} \right) = \Theta \left(\frac{1}{r k} \right), \quad b - b_k = \frac{1}{r} \left(\frac{1}{2} - s(\rho_k) \right) = O \left(\frac{1}{r w_k^2} \right) = O \left(\frac{1}{r k^2} \right).$$

Thus there exist k_1 and constants $0 < c < C < \infty$ such that for all $k \geq k_1$,

$$a \Delta_{k-1} + \frac{c}{r k} \leq \Delta_k \leq a \Delta_{k-1} + \frac{C}{r k}.$$

Iterating these comparison recursions yields $c_\Delta / (rk) \leq \Delta_k \leq C_\Delta / (rk)$ for all $k \geq k_1$, completing the proof. $\blacksquare$

D.8. Remark: proxy lower bounds vs positivity (smallest eigenvalue)

Notation. Let $\widehat{K}$ denote the empirical NTK Gram matrix at initialization, with entries $\widehat{K}_{ij} = \langle \nabla_\theta f(x_i), \nabla_\theta f(x_j) \rangle$. Let K denote the mean (deterministic) kernel Gram matrix with entries $K_{ij} = \mathbb{E}[\widehat{K}_{ij}]$. The *centered* spectrum refers to the restriction of these matrices to the subspace $\mathbf{1}^\perp$ of vectors orthogonal to the constant vector $\mathbf{1}$; we write $\widehat{K}|_{\mathbf{1}^\perp}$ and $K|_{\mathbf{1}^\perp}$ for these restrictions.

Remark 37 (Can the smallest eigenvalue become negative?) *No: $\widehat{K}$ is a Gram matrix of gradients, hence $\widehat{K} \succeq 0$ deterministically and all its eigenvalues are ≥ 0 . Restricting to the mean-zero subspace preserves positive semidefiniteness: for any $v \perp \mathbf{1}$, one has $v^\top \widehat{K} v \geq 0$, so $\widehat{K}|_{\mathbf{1}^\perp} \succeq 0$.*

What can happen is that a deterministic proxy lower bound on the centered eigenvalue scale is smaller than the magnitude of random fluctuations. Weyl’s inequality gives

$$\lambda_{\min}(\widehat{K}|_{\mathbf{1}^\perp}) \geq \lambda_{\min}(K|_{\mathbf{1}^\perp}) - \|\widehat{K} - K\|_2,$$

so if $\|\widehat{K} - K\|_2$ dominates the deterministic scale of $\lambda_{\min}(K|_{\mathbf{1}^\perp})$, the bound becomes vacuous (it may be negative), but the true eigenvalue still satisfies $\lambda_{\min}(\widehat{K}|_{\mathbf{1}^\perp}) \geq 0$. Thus, the relevant concern is not “going below zero” but rather “being driven close to zero” when the $1/(rL)$ signal is swamped by noise.

Fisher–Kibble decoupling (used for concentration). Fluctuations of $\widehat{K}$ around K are controlled by the bottleneck dimension r ; the joint law of the rank- r random projections is given by Fisher–Kibble decoupling (Lemma 9 in the main text). Let x, y be input vectors and x_1, y_1 their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#); [Hotelling \(1953\)](#), centered at the true correlation ρ .

- (u, v) follow Kibble’s [Kibble \(1941\)](#) bivariate Gamma (chi-square) law.
- Angular and radial parts are independent: $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

This decoupling is used in Section 5 and in the concentration bounds (Theorem 27); see Appendix C.2 for explicit densities.

D.9. Remark: infinite-depth limit after centering

Remark 38 (Infinite-depth limit: existence and degeneracy after centering) *The correlation recursion satisfies $1 - \rho_k = O(k^{-2})$ as in full-width MLPs at the EOC [Terjék and González-Sánchez \(2025\)](#); [Hayou et al. \(2022\)](#), so deep features become increasingly aligned. For the mean RF-LR NTK recursion (83), Theorem 5 implies that for any fixed $\rho_1 \in (-1, 1)$,*

$$\Theta^{(L)}(\rho_1) \rightarrow \Theta_\star(r) \quad \text{and} \quad \Theta^{(L)}(1) \rightarrow \Theta_\star(r),$$

so the entrywise infinite-depth limit of the (mean) Gram matrix is a constant kernel: $K^{(L)} \rightarrow \Theta_\star(r) \mathbf{1}\mathbf{1}^\top$. Consequently, after centering (restricting to the orthogonal complement of the constant mode), the relevant part of $K^{(L)}$ tends to zero and the spectrum collapses. In particular, a non-degenerate deep limit on $\mathbf{1}^\perp$ can only hold under a joint scaling (e.g. r growing with L) or after an explicit rescaling of the centered operator (equivalently, rescaling time/learning rate in kernel gradient descent).